SUSTech Southern University of Science and Technology

EE104 Fundamentals of Electric Circuits (Lesson 06)

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Review: Steady-sate DC circuits (a)

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- Mid-term exam

Chapters: chapters 1-7 **Time:** Nov.9th Saturday **Exam:** closed-book in English.

Location: The 3rd teaching building, and rooms can be found in the table below.

Tools: answer sheets and stretch paper will be provided. You will need calculators, pens and student ID cards. No phones, no smart watches, no books or any other documents related to circuits.

Key info. 1: One A4 cheating paper is allowed (both side, hand writing or printing).

Key info. 2: No make-up or online exams.

Key info. 3: SUBMIT all the exam paper, answer sheets, scratch papers, your cheating paper. **IT IS YOUR DUTY** to make sure all these documents are safely submitted to the TA at the end of the exam. IF a student loses one of these documents, he will be considered as **CHEATING** and score will be **ZERO**.

Class	1	2	3	4
Rooms	Rm105	Rm. 106	Rm. 107	Rm. 108
教室	三教105	三教106	三教107	三教108

Review: Steady-sate DC circuits (a)

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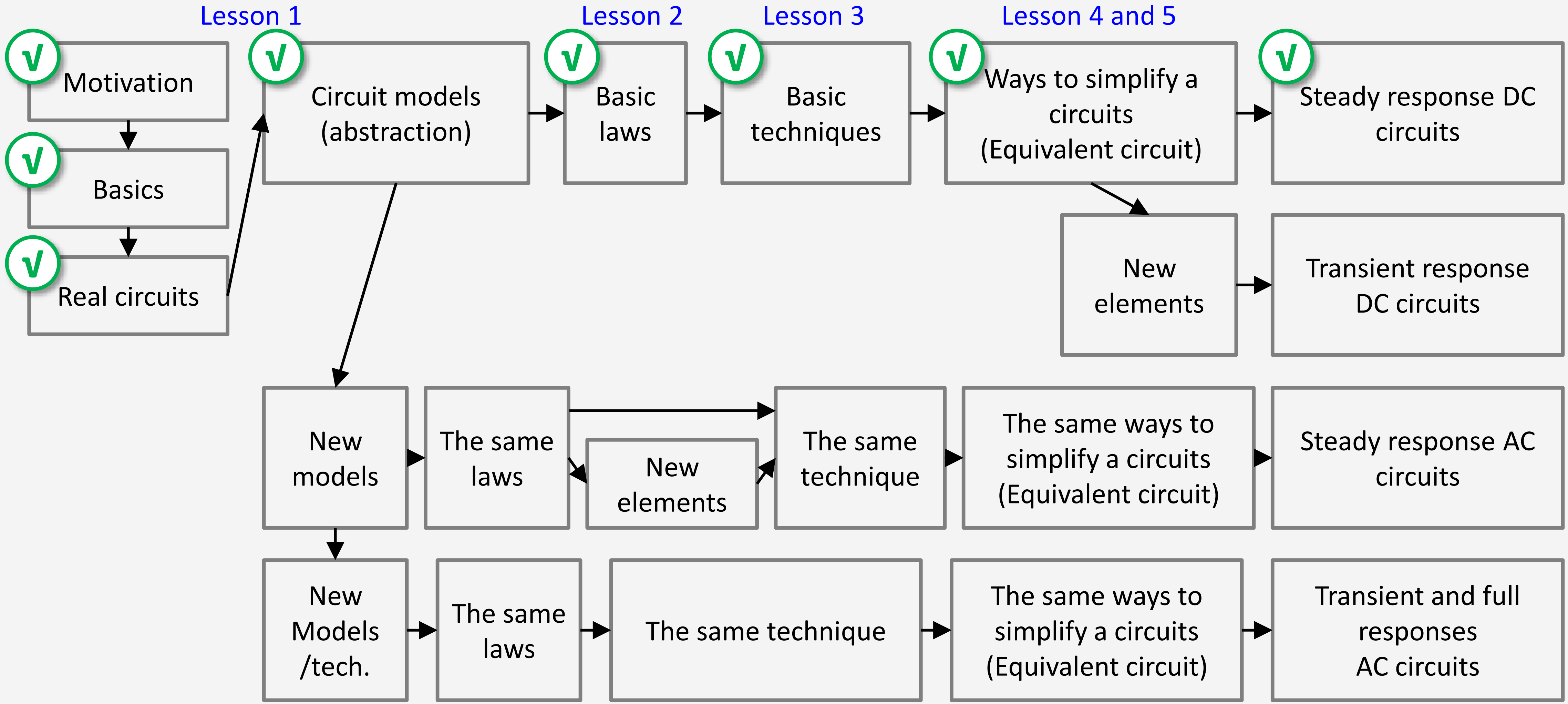
- EE104 road map

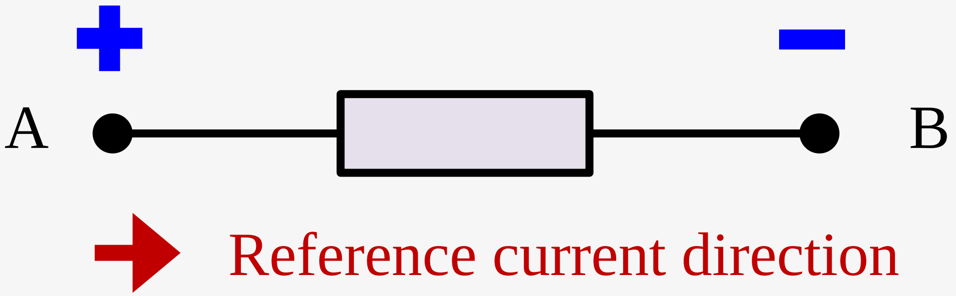


Review: Steady-sate DC circuits (b)

- Lesson 1: reference directions

Associated reference directions

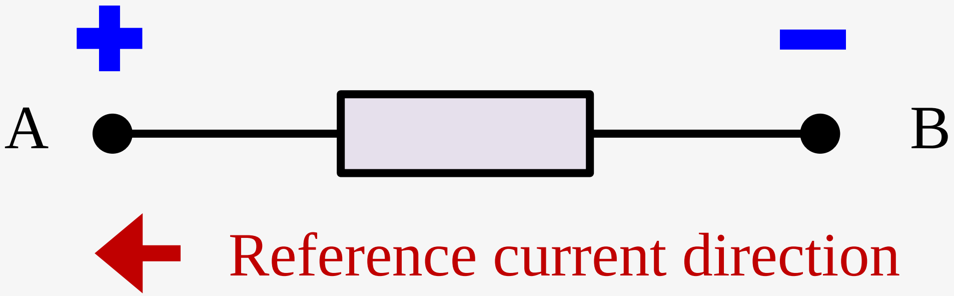
Reference voltage direction



Reference current direction

Nonassociated reference directions

Reference voltage direction



Reference current direction

$P = ui > 0$ absorb energy
$P = ui < 0$ supply energy

$P = ui > 0$ supply energy
$P = ui < 0$ absorb energy

Review: Steady-sate DC circuits (c)

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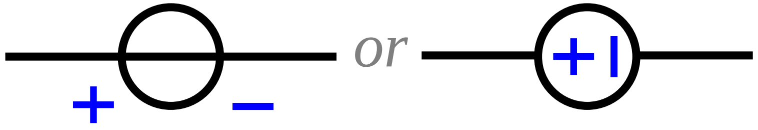
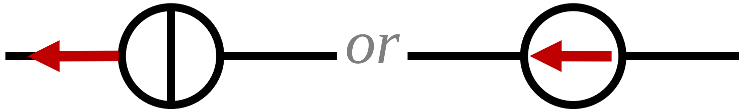
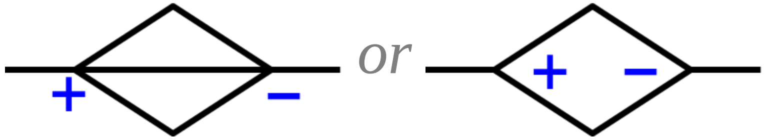
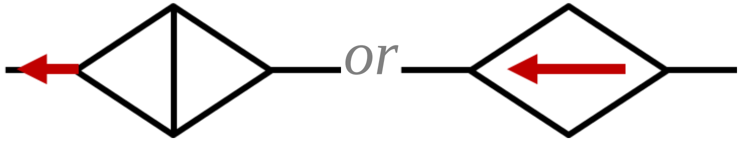
- Lesson 1: dependent sources

	Voltage sources	Current sources
Independent	 V_{out} V_{out}	 I_{out} I_{out}
Dependent	 $V_{out} = kV_{in}$ (or kI_{in}) $V_{out} = kV_{in}$ (or ki_{in})	 $I_{out} = kI_{in}$ (or kV_{in}) $I_{out} = kI_{in}$ (or kV_{in})

Review: Steady-sate DC circuits (d)

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- Lesson 2: The circuits big three

Ohm's law	Ohm's law states that the voltage v across a conductor is directly proportional to the current i flowing through it, provided all physical conditions and temperature remain constant.
KCL	The algebraic sum of currents entering a node (or a closed boundary is zero. In other words, the sum of the currents entering a node is equal to that leaving the node.
KVL	The algebraic sum of all voltages around a loop is zero, or Sum of voltage drops = Sum of voltage rises, which is based on the conservation of energy.

Review: Steady-sate DC circuits (e)

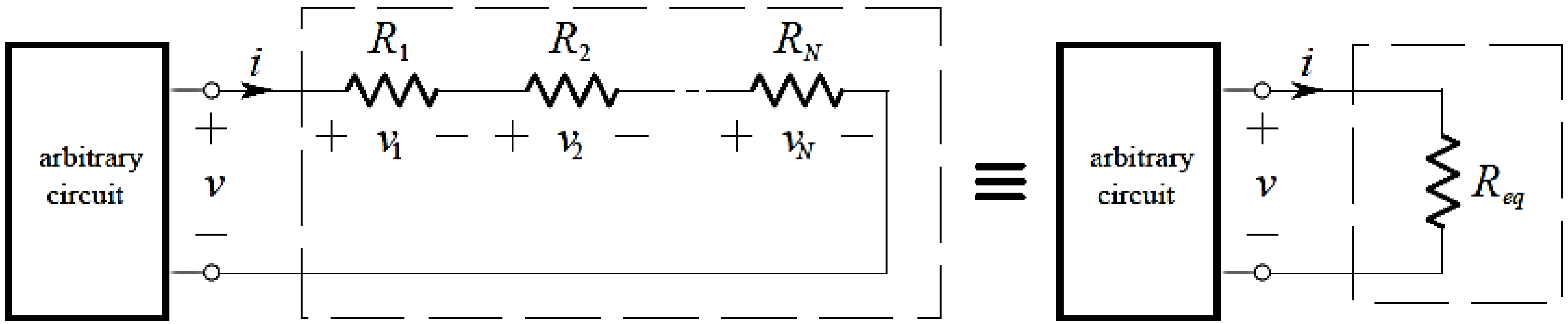
- Lesson 2: equivalent circuits + voltage division + current division

a) An equivalent circuit is a simplified model of an existing circuit, used to simplify the analysis.

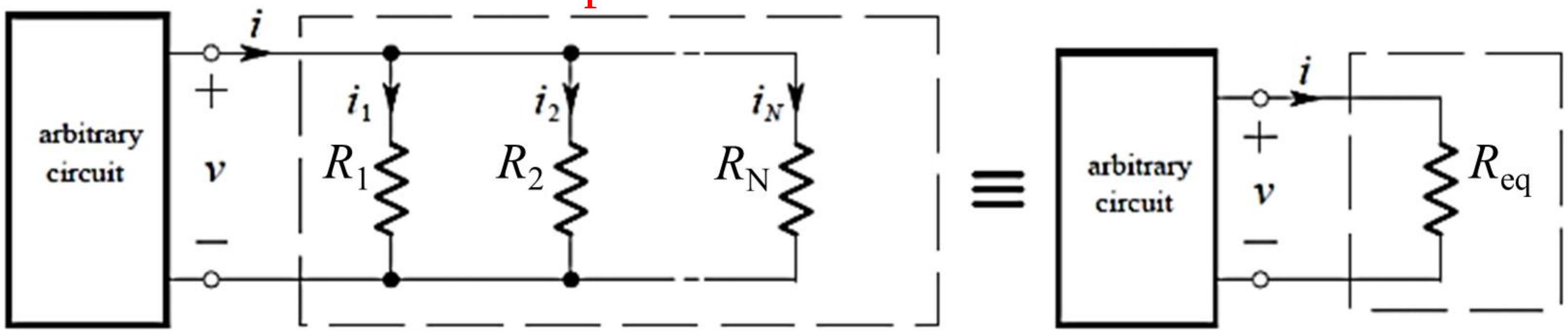
b) Two circuits are said to be equivalent if they have identical *v*-*i* characteristics at the ports.

c) It is of vast importance to understand the concept of equivalent circuits.

Resistors in series



Resistors in parallel



Review: Steady-sate DC circuits (f)

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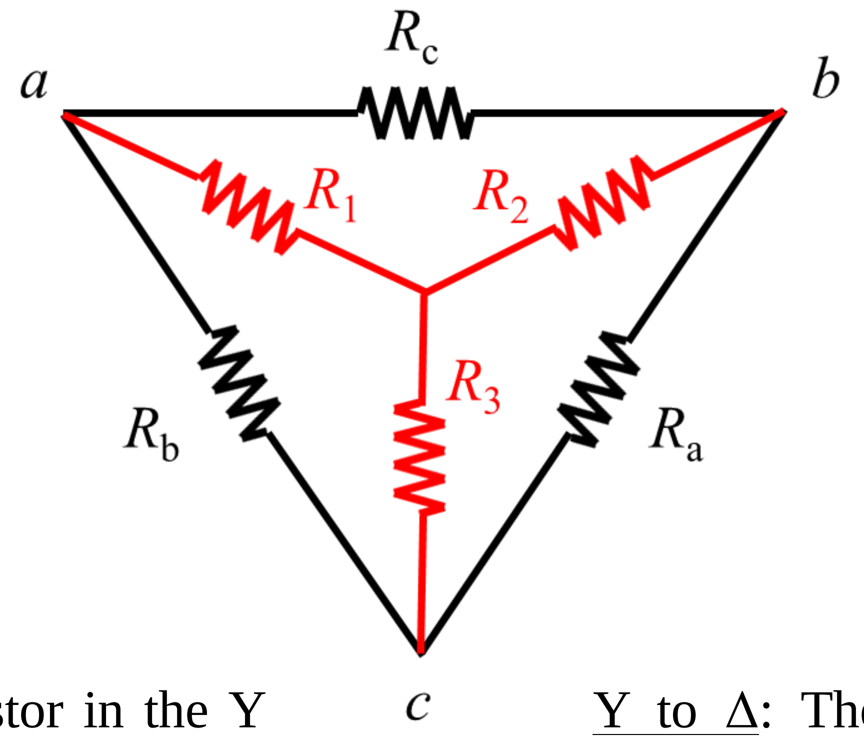
- Lesson 2: Y-Δ and Δ-Y transformations

Δ to Y

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_b R_a}{R_a + R_b + R_c}$$



Y to Δ

$$R_a = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_1}$$

$$R_b = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_2}$$

$$R_c = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_3}$$

Δ to Y: the resistance of a resistor in the Y network is the product of the resistance of the two adjacent resistors in the Δ network, divided by the sum of the resistance of the three Δ resistors.

Y to Δ: The resistance of a resistor in the Δ network is the sum of all possible products the resistance of the Y resistors taken two at a time, divided by the resistance of the opposite Y resistor.

Review: Steady-sate DC circuits (g)

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- Lesson 3: Nodal analysis

Step 01
Label the Nodes

a) Pick any node as the voltage reference. Label its voltage as 0 V. Label dependent sources with v_s or i_s.
b) If any voltage sources are connected to a labelled node, label their other ends by adding the value of the source onto the voltage of the labelled end. Repeat as many times as possible.
c) Pick an unlabelled node and label it, then loop back to step b) until all nodes are labelled.
d) For each dependent source, write down an equation that expresses its value in terms of other node voltages.

Step 02
KCL at nodes

e) Write down a KCL equation for each "normal" node (i.e. one that is not connected to a floating voltage source).
f) Write down a KCL equation for each "super-node". A super-node consists of a set of nodes that are joined by floating voltage sources and includes any other components joining these nodes.

Step 03
Ohm's Law

g) Substitute the currents using Ohm's Law

Step 04
Solve equations

h) Solve the equation

Review: Steady-sate DC circuits (h)

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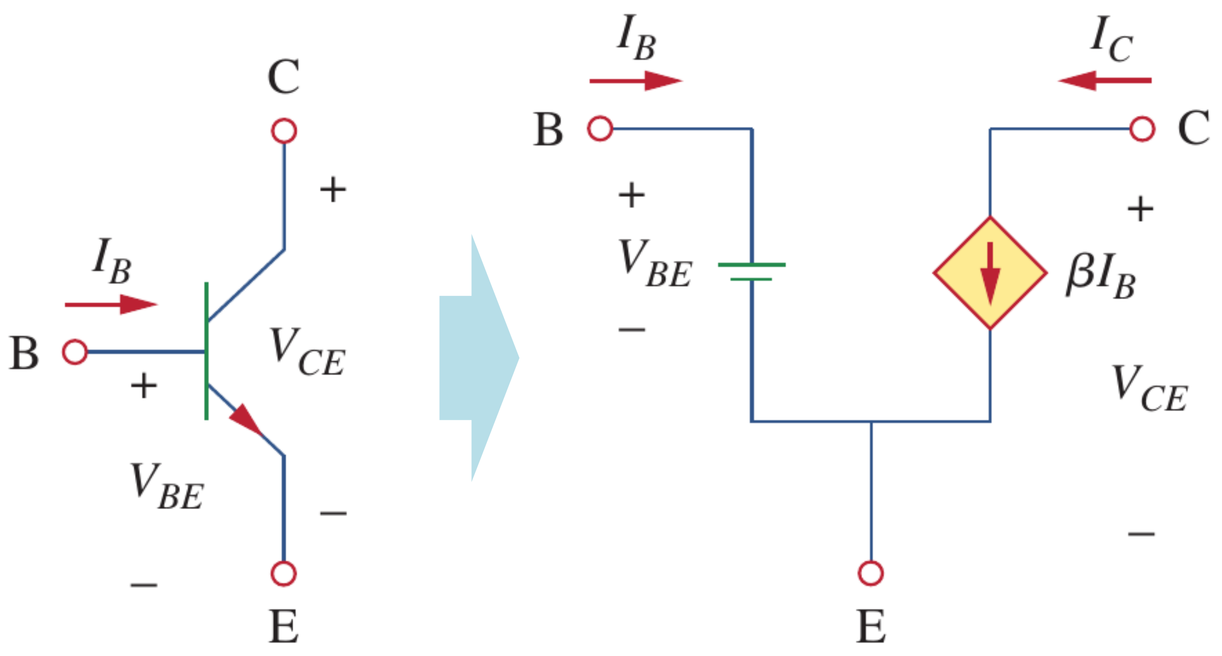
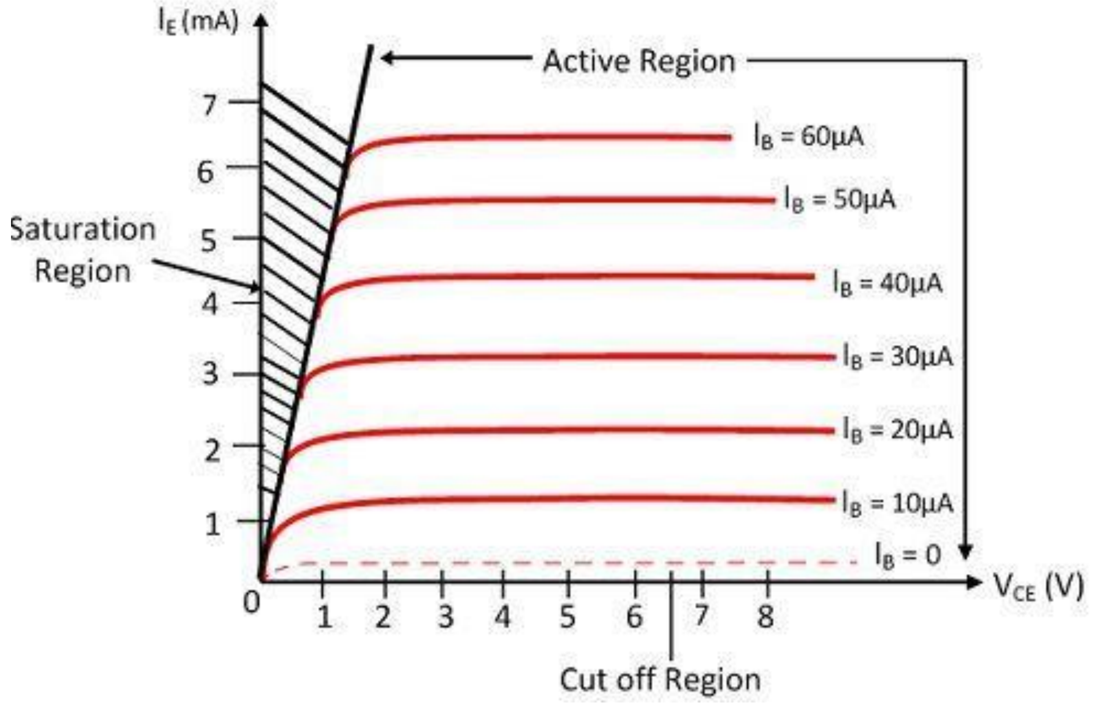
- Lesson 3: Mesh analysis

Step 01 Label the Mesh	◻ Identify the mesh and assign a mesh current to each of them ◻ Select the same direction for all mesh currents (i.e. clockwise)
Step 02 KVL for meshes	◻ Apply KVL for each mesh based on reference voltage directions ◻ Each mesh results in an equation in terms of mesh currents
Step 03 Ohm’s Law	◻ Apply Ohm’s Law to express each voltage ◻ Pay extra attention to the reference voltage directions
Step 04 Solve equations	◻ Solve the equations obtained in previous steps. ◻ Number of equations will be equal to the number of meshes

Review: Steady-sate DC circuits (j)

- Lesson 3: DC transistor circuits (equivalent circuits)





$$I_E = I_B + I_C, \quad I_E = (1+\beta)\cdot I_B$$

Review: Steady-sate DC circuits (k)

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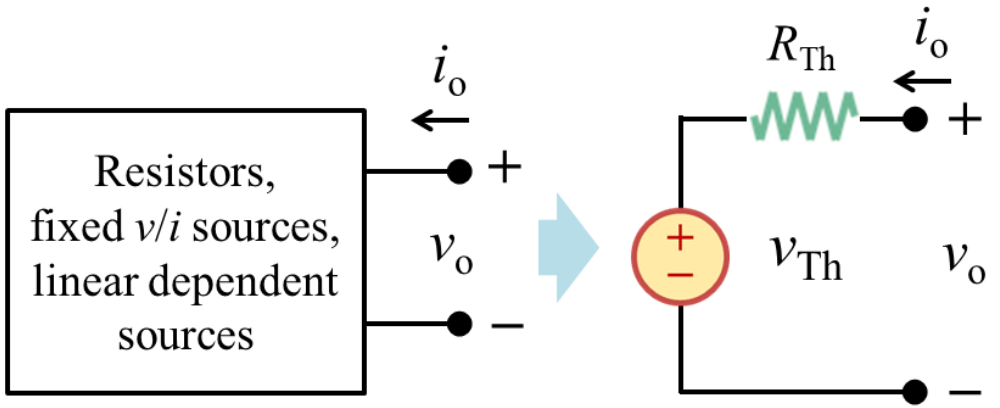
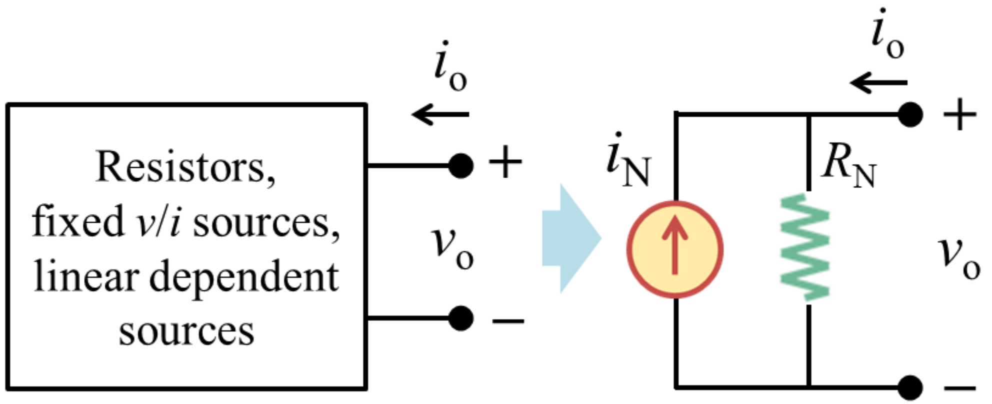
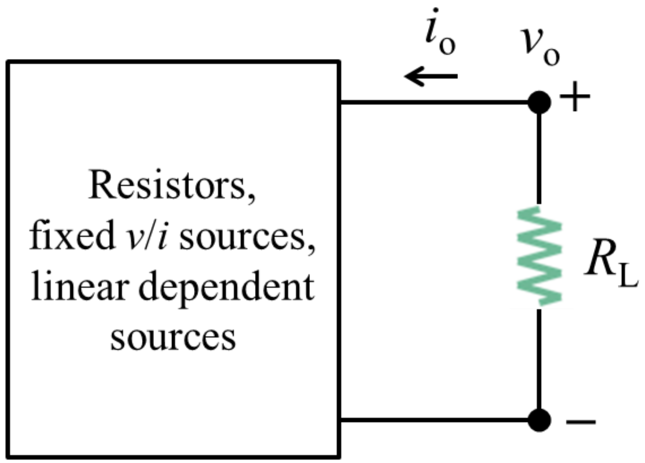
- Lesson 4: Thevenin and Norton theorems + source transformation

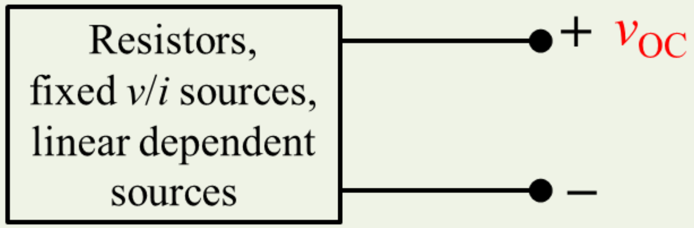
Thévenin's theorem	**Norton's theorem**
	
$v_o = R_{Th} \cdot i_o + V_{Th}$	$i_o = v_o / R_N - i_N$
Open-circuit voltage: $v_{OC} = v_{Th}$	Open-circuit voltage: $v_{OC} = i_N \cdot R_N$
Short-circuit current: $i_{SC} = v_{Th}/R_{Th} = -i_o$	Short-circuit current: $i_{SC} = i_N = -i_o$
Equivalent R: $R_{Th} = v_{OC}/i_{SC} = R_{eq}$	Equivalent R: $R_N = v_{OC}/i_{SC} = R_{eq}$

Thévenin-Norton transformation: $v_{Th} = i_N \cdot R_{Th}$

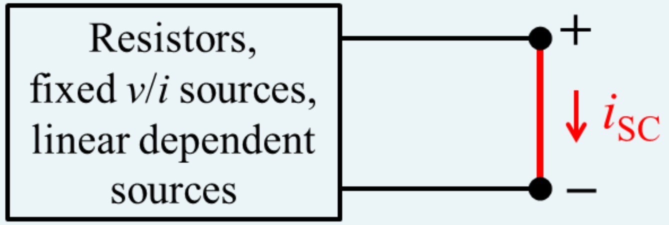
Review: Steady-sate DC circuits (I)

- Lesson 4: v_{oc}, i_{sc} and R_{eq}

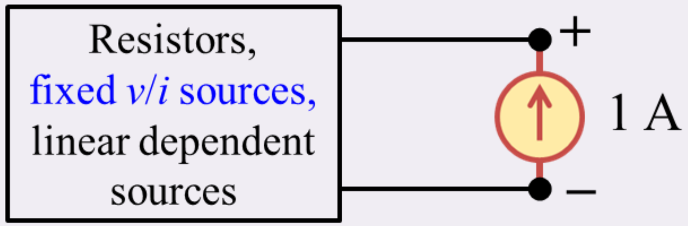




To find v_{OC}, remove the load and calculate the v_{OC} using nodal/mesh analysis.



To find i_{SC}, replace the load with an ideal wire and calculate the i_{SC} using nodal/mesh analysis.



To find R_{eq}, replace the load with a 1 A current source (i_o = 1 A), turn off the independent sources, calculate the v_O, and $R_{eq} = v_O/i_O$.

*You can also use a 1 V voltage source

Review: Steady-sate DC circuits (m)

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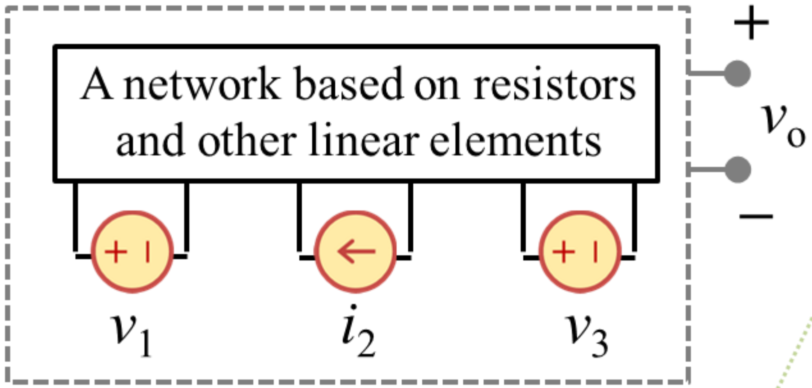
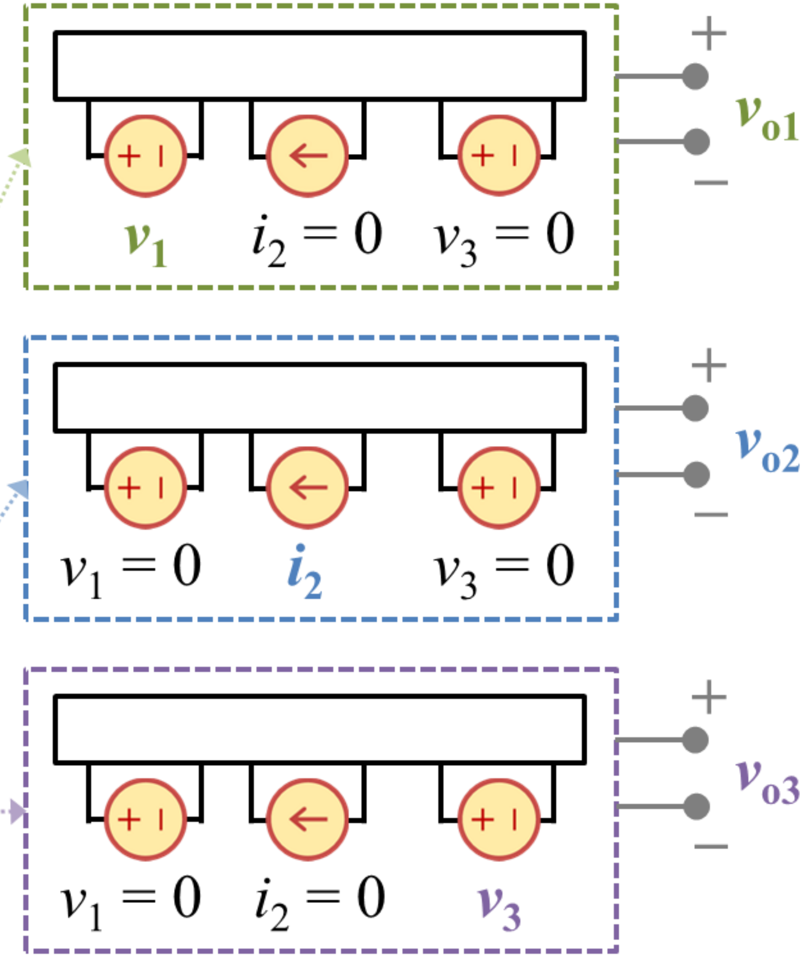
- Lesson 4: the maximum power transfer

In a linear system, the superposition principle (叠加原理) states that the voltage across (or current through) an element is the algebraic sum of those caused by each independent source acting alone (每个独立源单独作用).



$$
\begin{aligned}
v_o &= Av_1 + Bi_2 + Cv_3 \\
&= \mathbf{Av_1} + B\times 0 + C\times 0 \\
&\quad + A\times 0 + \mathbf{Bi_2} + C\times 0 \\
&\quad + A\times 0 + B\times 0 + \mathbf{Cv_3} \\
&= \mathbf{v_{o1}} + \mathbf{v_{o2}} + \mathbf{v_{o3}}
\end{aligned}
$$



Review: Steady-sate DC circuits (n)

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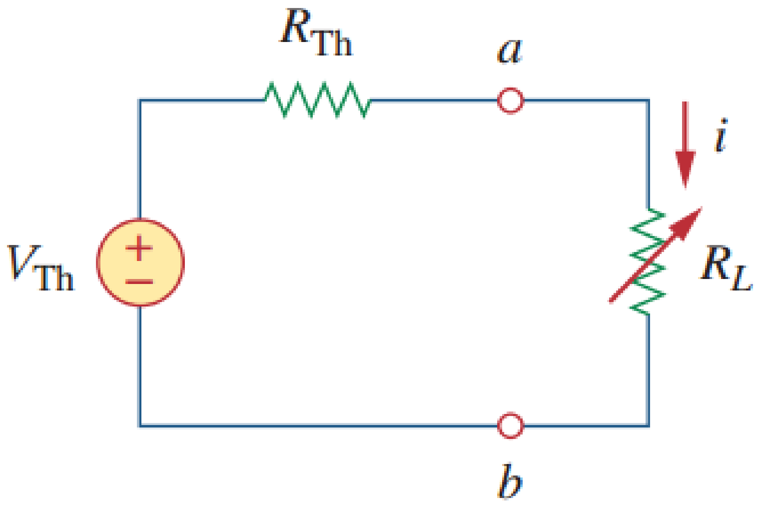
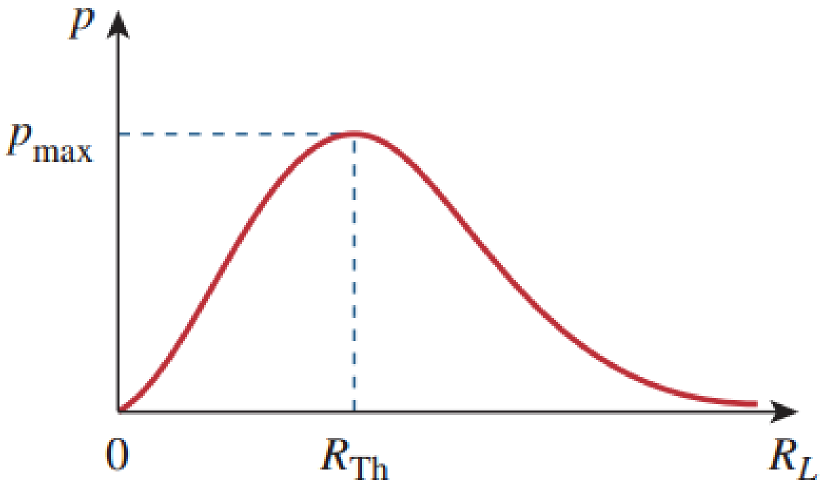
- Lesson 4: superposition theorem

$$R_L = R_{\text{Th}} \qquad p_{\max} = \frac{V_{\text{Th}}^2}{4R_{\text{Th}}}$$



$$p = i^2 R_L = \left(\frac{V_{\text{Th}}}{R_{\text{Th}} + R_L}\right)^2 R_L$$

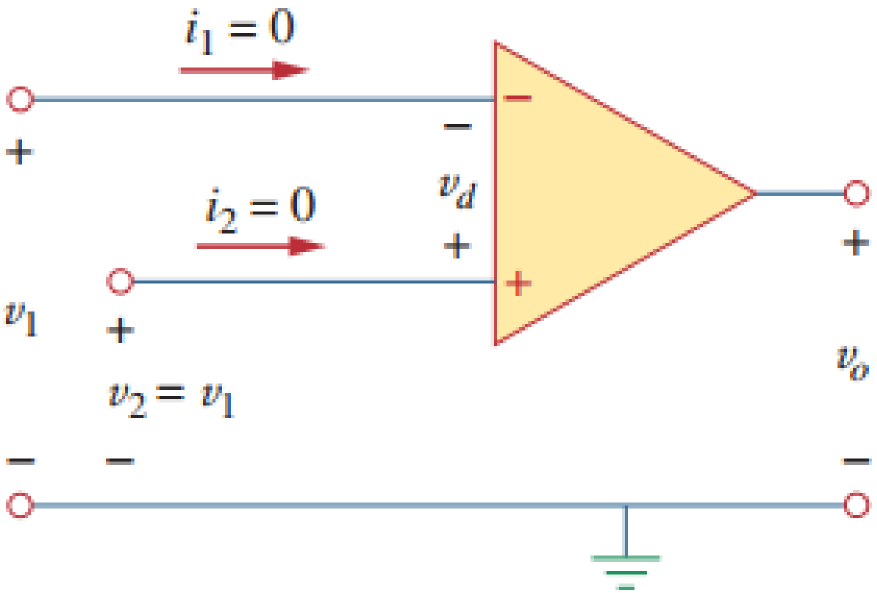


a) Maximum power transfer theorem states that maximum power is transferred to the load when the load resistance equals to the Thevenin resistance of the circuit seen by the load ($R_{\text{L}} = R_{\text{Th}}$).

Review: Steady-sate DC circuits (o)

- Lesson 5: DC op-amp circuits

Ideal op-amps	A	R_i	R_o
	∞	∞	0



For ideal op-amps:

a) To calculate voltages, the two input terminals can be considered as "shorted".

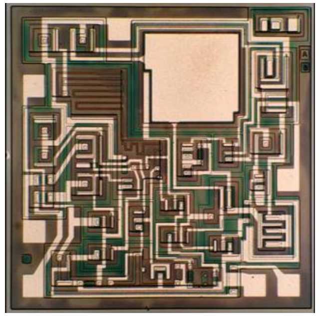
$$v_2 - v_1 = 0$$

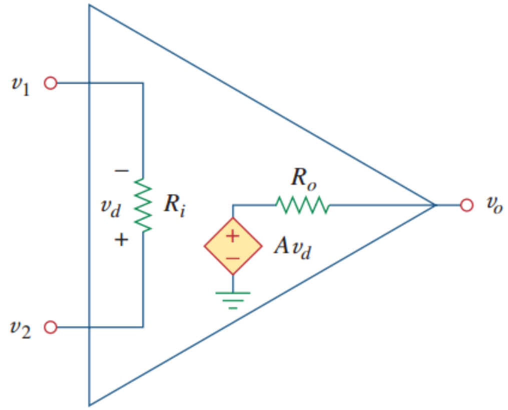
b) To calculate currents, the two input terminals and the internal portion of the op-amp can be considered as "open".

$$i_1 = i_2 = 0$$

Review: Steady-sate DC circuits (q)

- Lesson 5: DC op-amp circuits





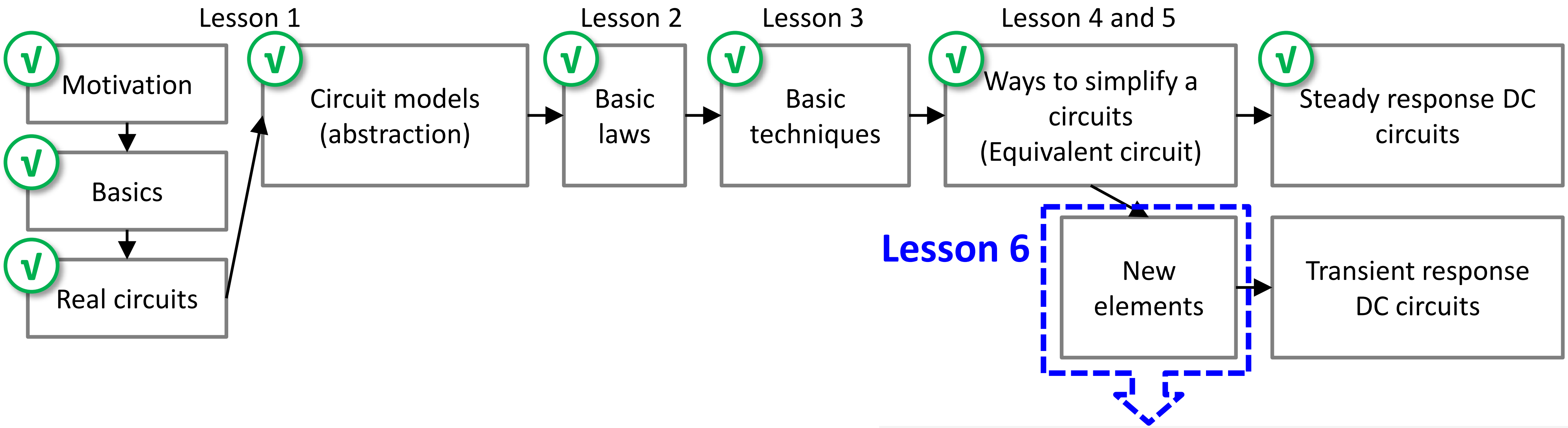
The equivalent circuit

The golden rules: 虚短 & 虚开

1. Inverting op-amp circuit
2. Non-inverting op-amp circuit
3. Summing op-amp circuit
4. Difference op-amp circuit
5. Cascaded op-amp circuit

Abstract: lesson 6

- EE104 road map



1. Capacitors
2. Inductors
3. Integrators
4. Differentiators

Lesson 6 Capacitors and inductors

6.1 Capacitor basics (a)

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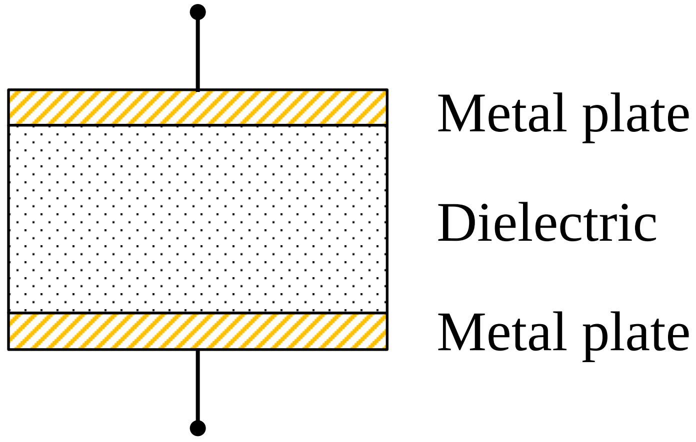
- Capacitors:

a) A capacitor is a passive element that stores energy in the electric field.

b) A capacitor consists of two conducting layers separated by a layer of dielectric insulator.

c) Besides resistors, capacitors are the most common electrical components.



Polyester capacitor *Ceramic capacitor* *Electrolytic capacitor* *Trimmer capacitor* *Filmtrim capacitor*

6.1 Capacitor basics (b)

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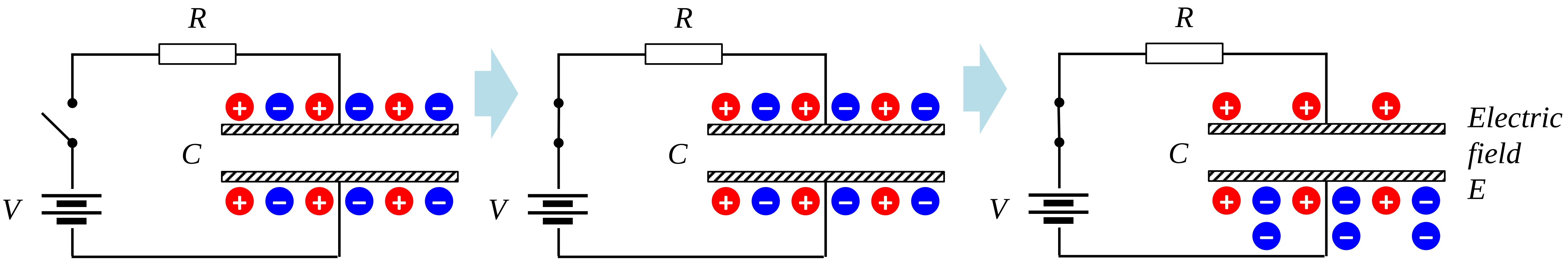
- Capacitance:



a) If a voltage is applied, changes will accumulate on one metal plate, and balancing charges with the opposite polarity will be present on the other plate to preserve charge neutrality.

b) 电容器 = "电荷的容器": $q = Cv$.

c) C - capacitance, the ratio of the net charge on one plate of a capacitor to the voltage difference between the two plates, commonly measured in picofarad (pF) and microfarad (μF). 1 farad (法拉) = 1 coulomb/volt.

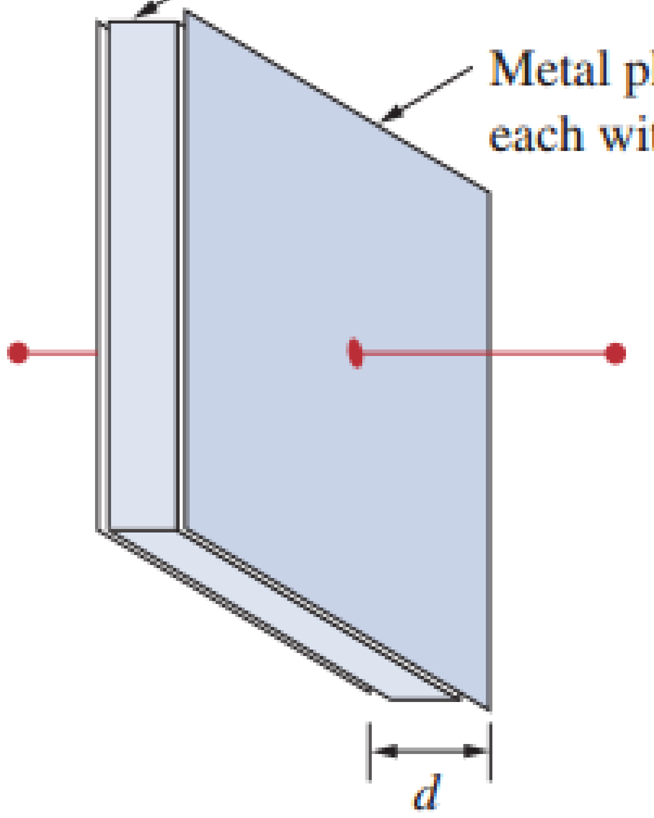
6.1 Capacitor basics (c)

- Capacitance:

a) For a planar type capacitor, its capacitance is given by the following equation, in which ε is the permittivity (介电常数，电容率) of the dielectric material, *A* is the area of a metal plate, and *d* is the thickness of the dielectric material.

$$\boxed{C = \frac{\varepsilon A}{d}}$$

b) Permittivity is the ability of a material to store electrical potential energy under the influence of an electric field. The relative permittivity of a material is its permittivity expressed as a ratio relative to the vacuum permittivity.



6.1 Capacitor basics (d)

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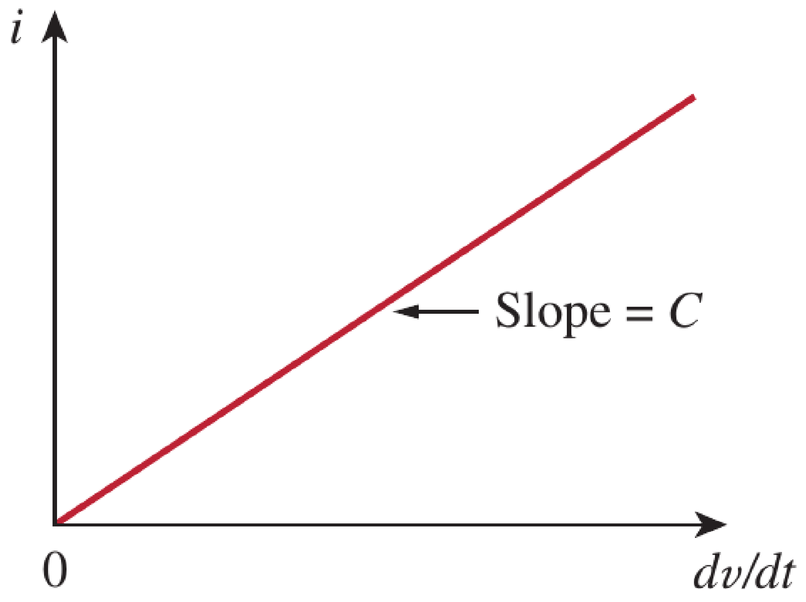
- *i-v* characteristics:

a) The *i-v* characteristics of an ideal capacitor is shown in the right figure.

$$\begin{cases} i = dq/dt \\ q = Cv \end{cases} \Rightarrow \boxed{i = C\frac{dv}{dt}}$$



b) And *v-i* characteristics:

$$\boxed{v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)d\tau = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)}$$

which shows that the capacitor voltage depends on the past history of the capacitor current. Hence, the capacitor has memory—a property that is often exploited.

6.1 Capacitor basics (e)

- Capacitors are linear elements:

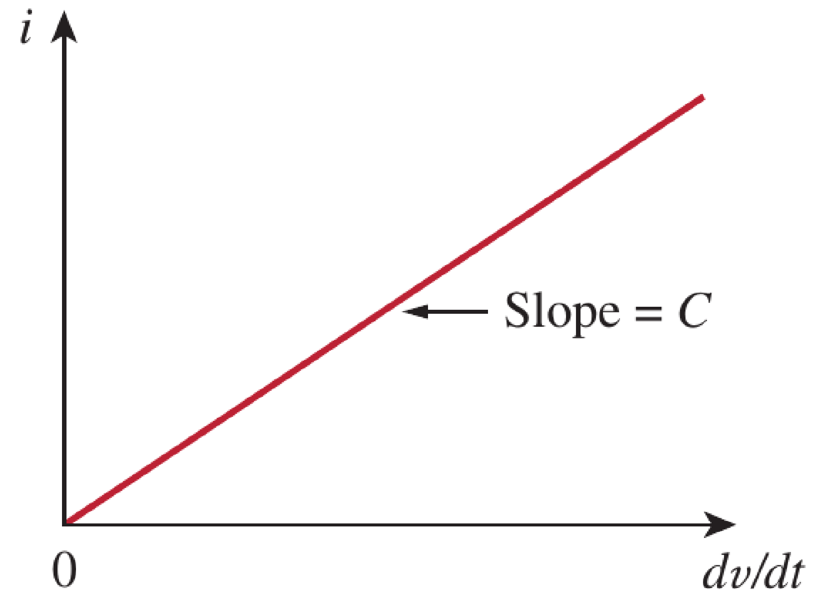
a) Capacitors are linear elements. $i(t) = C\dfrac{dv(t)}{dt}$

b) Recall Lesson 4:

Homogeneity? Yes: $C\dfrac{d(av(t))}{dt} = a \cdot C\dfrac{dv(t)}{dt}$

Additivity? Yes

$$C\frac{d(v_1(t) + v_2(t))}{dt} = C\frac{dv_1(t)}{dt} + C\frac{dv_2(t)}{dt}$$



6.1 Capacitor basics (f)

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- Power and energy:

a) Instant power (瞬态功率): $p = vi = Cv\dfrac{dv}{dt}$

b) The energy stored in the capacitor:

$$w = \int_{-\infty}^{t} v'i'd\tau = C\int_{-\infty}^{t} v'\frac{dv'}{d\tau}d\tau = C\int_{0}^{v} v'dv' \boxed{= \frac{1}{2}Cv^2 = \frac{q^2}{2C}}$$

in which $v(-\infty) = 0$, because the capacitor was uncharged at $t = -\infty$.

This equation represents the energy stored in the electric field that exists between the plates of the capacitor. This energy can be retrieved, since an ideal capacitor cannot dissipate energy.

6.1 Capacitor basics (g)

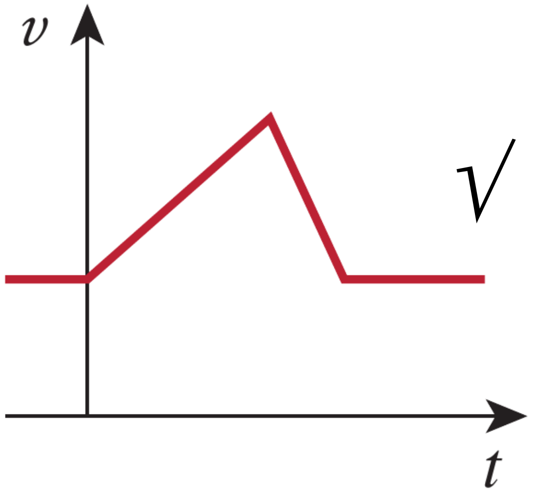
- Other important properties of capacitors:

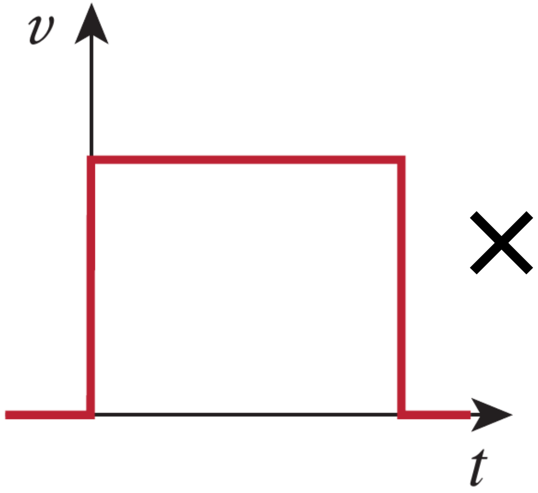
a) A capacitor is an open circuit to DC ($i = 0$):

$$i = C\frac{dv}{dt}, \text{and } \frac{dv}{dt} = 0 \text{ for DC}$$

b) The voltage on a capacitor cannot change abruptly:

The capacitor resists an abrupt change in the voltage across it. A discontinuous change in voltage requires an infinite current, which is physically impossible.





6.1 Capacitor basics (h)

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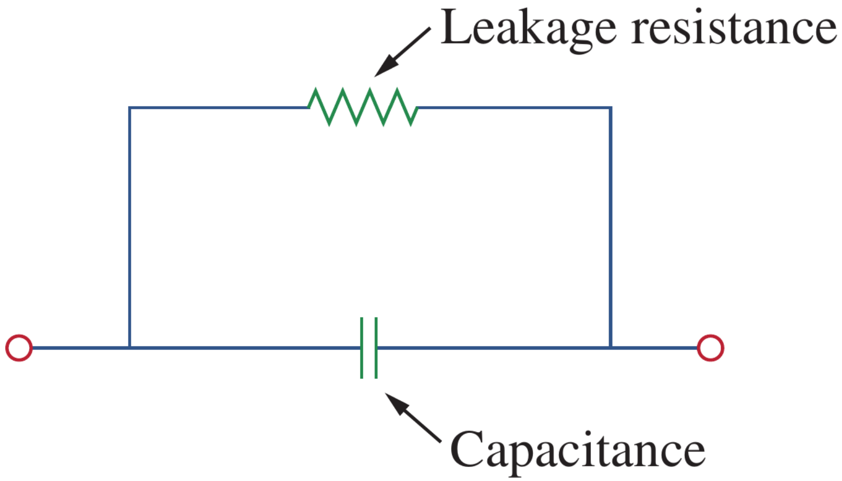
- Ideal and nonideal capacitors

a) The ideal capacitor does not dissipate energy. It takes power from the circuit when storing energy in its field and returns previously stored energy when delivering power to the circuit.

b) A real, nonideal capacitor has a parallel-model leakage resistance and could be nonlinear. The leakage resistance may be as high as 100 MΩ (can be neglected for most practical applications).



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- Example 6.01

Example 6.01: (a) calculate the charge stored on a 3 pF capacitor with 20 V across it, and find the energy stored in the capacitor. (b) If the voltage across the capacitor is *v*(t) = 10 cos(6000t) V, calculate the current *i*.

Solution: (a) $q = Cv = 3 \times 10^{-12} \times 20 = 60 \times 10^{-12}\ \mathrm{F} = 60\ \mathrm{pC}$

$$w = \frac{1}{2}Cv^2 = \frac{1}{2} \times 3 \times 10^{-12} \times 400 = 600\ \mathrm{pJ}$$

(b) $$i = C\frac{dv}{dt} = C\frac{d(10cos6000t)}{dt} = -3 \times 10^{-12} \times 6000 \times 10\sin(6000t)$$

$$= -1.8 \times 10^{-7} \sin(6000t)\ \mathrm{A}$$

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- Example 6.02

Example 6.02: Determine the voltage across a 2-μF capacitor if the current through it is

$$i(t) = 6e^{-3000t} \text{ mA}$$

Assume that the initial voltage is zero.

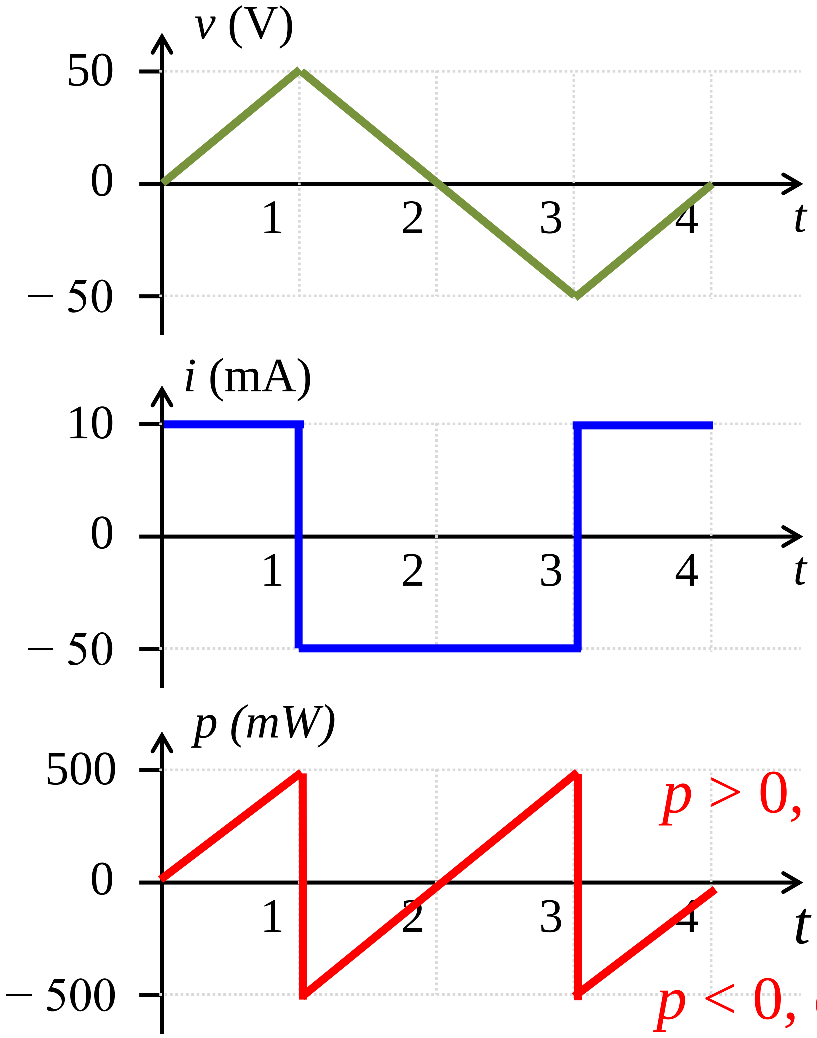
Solution: Since $v = \frac{1}{C}\int_0^t i\,dt + v(0)$ and $v(0) = 0$,

$$v = \frac{1}{2\times 10^{-6}}\int_0^t 6e^{-3000t}\,dt \cdot 10^{-3}$$

$$= \frac{3\times 10^3}{-3000}e^{-3000t}\Big|_0^t = (1 - e^{-3000t}) \text{ V}$$

6.1 Capacitor basics (j)

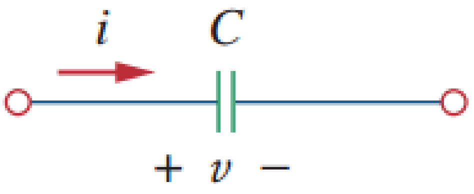
- Example 6.03



$p > 0$, charging

$p < 0$, discharging

Example 6.03: Determine the current through a 200-μF capacitor when a voltage as shown in the left figure is applied; also calculate the power on the capacitor.



$$v(t) = \begin{cases} 50t \text{ V} & 0 < t < 1 \\ 100 - 50t \text{ V} & 1 < t < 3 \\ -200 + 50t \text{ V} & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Solution:

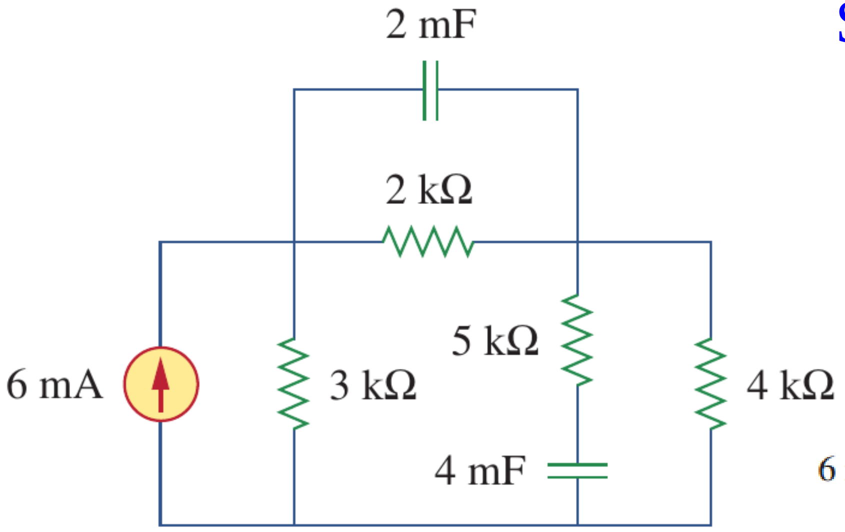
(1) $0 < t < 1$, $dv/dt = 50$, $i = 0.2\text{ m} \times 50 = 10$ mA

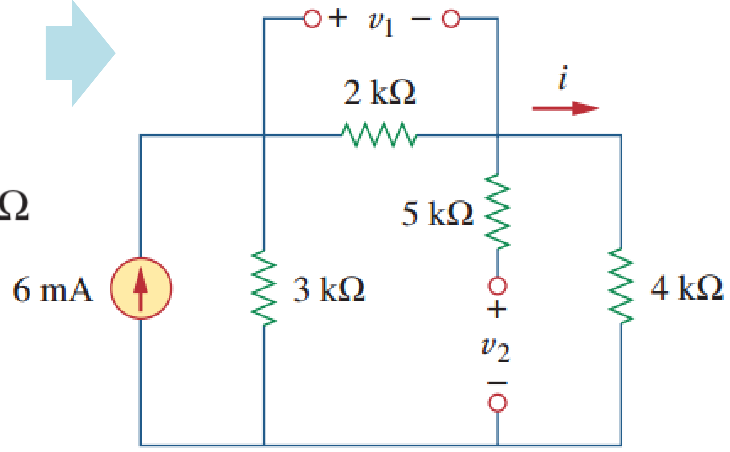
(2) $1 < t < 3$, $dv/dt = -50$, $i = -10$ mA

(3) $3 < t < 4$, $dv/dt = 50$, $i = 10$ mA

- Example 6.04

Example 6.04: Obtain the energy stored in each capacitor in figure below when it reaches a stable status.





Solution: capacitors appear as open-circuit

(1) Current across the 2 kΩ resistor:

$$i_1 = 6\text{ mA} \times 3\text{ k}/(3\text{k} + 2\text{k} + 4\text{k})$$
$$= 2\text{ mA (current division)}$$

(2) $v_1 = 2\text{ mA} \times 2\text{ k} = 2\text{ V}$

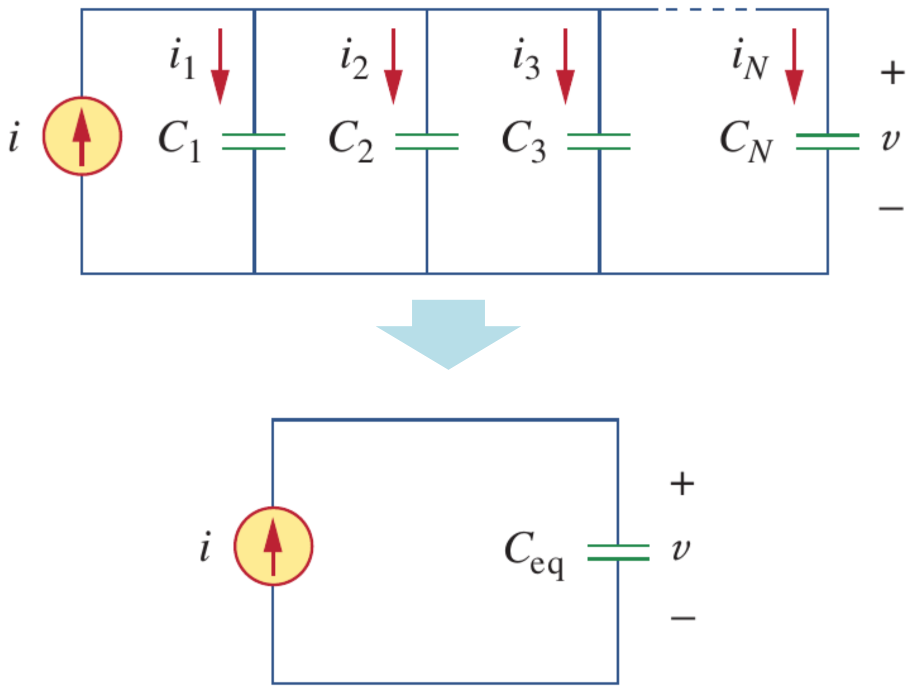
$v_2 = 2\text{ mA} \times 4\text{ k} = 8\text{ V}$

(3) $w_1 = C_1 v_1^2/2 = 16\text{ mJ}$

$w_2 = C_2 v_2^2/2 = 128\text{ mJ}$

6.2 Capacitor networks (a)

- Capacitors in parallel (并联)



a) Applying KCL at the top node:

$$i = i_1 + i_2 + i_3 + \cdots + i_N$$

b) Since $i = C dv/dt$, as:

$$i = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + C_3 \frac{dv}{dt} + \cdots + C_N \frac{dv}{dt} = \left(\sum_{k=1}^{N} C_k \right) \frac{dv}{dt} = C_{eq} \frac{dv}{dt}$$

c) For parallel capacitors, the equivalent capacitance is:

$$\boxed{C_{eq} = C_1 + C_2 + C_3 + \cdots + C_N}$$

The same formula as for the conductance G.

6.2 Capacitor networks (b)

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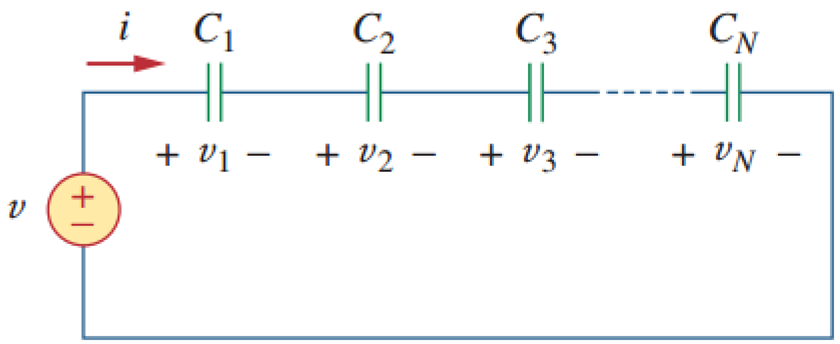
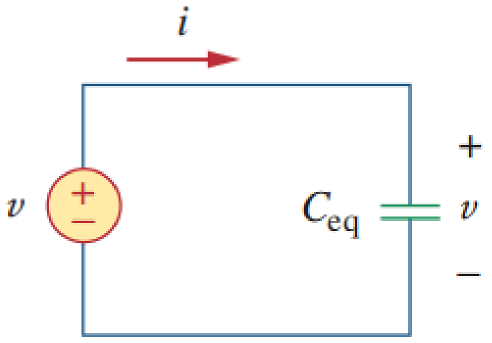
- Capacitors in series （串联）





a) Applying KVL for the loop: $v = v_1 + v_2 + v_3 + \cdots + v_N$

b) Since $v = \dfrac{1}{C}\displaystyle\int_{-\infty}^{t} i(\tau)d\tau$

$$v = \frac{1}{C_1}\int_{-\infty}^{t} i(\tau)d\tau + \frac{1}{C_2}\int_{-\infty}^{t} i(\tau)d\tau + \cdots + \frac{1}{C_N}\int_{-\infty}^{t} i(\tau)d\tau$$

$$= \left(\frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_N}\right)\int_{-\infty}^{t} i(\tau)d\tau$$

$$= \frac{1}{C_{eq}}\int_{-\infty}^{t} i(\tau)d\tau = \frac{1}{C_{eq}}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$$

c) So: $\boxed{\dfrac{1}{C_{eq}} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \dfrac{1}{C_3} + \cdots + \dfrac{1}{C_N}}$

6.2 Capacitor networks (c)

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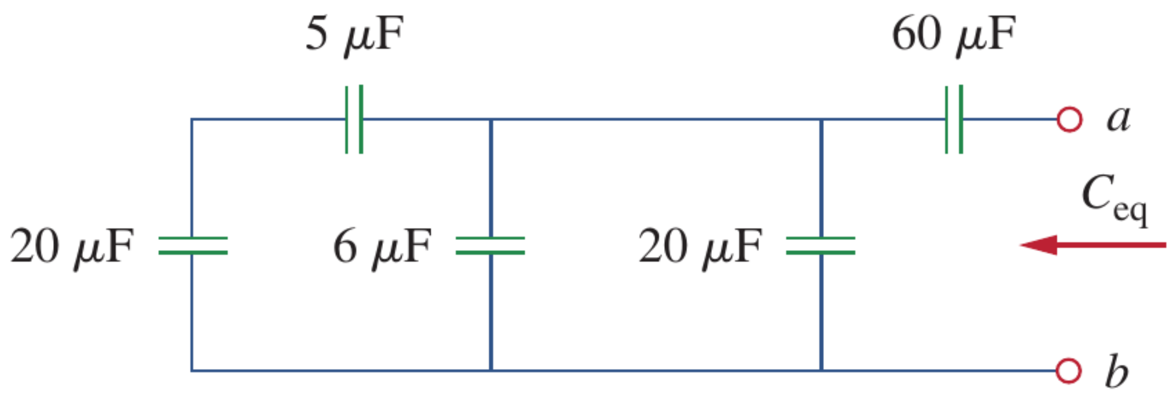
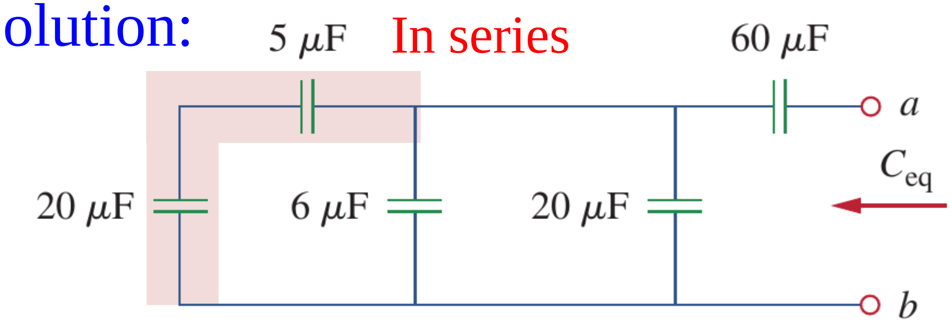
- Example 6.05

Example 6.05: Find the equivalent capacitance seen between terminals *a* and *b* of the following circuit.

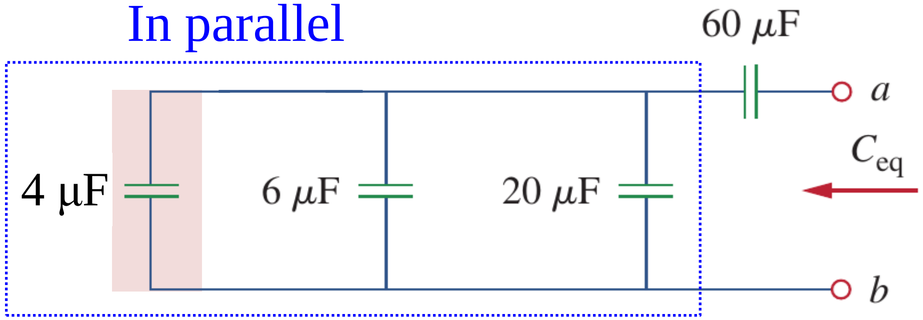


Solution:

In series

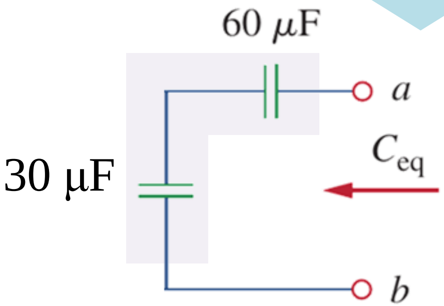


In parallel

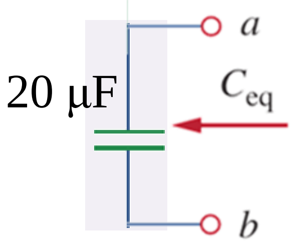


$$\frac{1}{C'_{eq}} = \frac{1}{20\mu} + \frac{1}{5\mu}$$

$$C'_{eq} = \frac{20 \times 5}{20 + 5} = 4\ \mu\text{F}$$



$$C''_{eq} = 4 + 6 + 20 = 30\ \mu\text{F}$$



$$\frac{1}{C_{eq}} = \frac{1}{30\mu} + \frac{1}{60\mu}$$

$$C_{eq} = \frac{60 \times 30}{60 + 30} = 20\ \mu\text{F}$$

6.2 Capacitor networks (d)

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- Example 6.05

Example 6.05: Find the voltage across each capacitor when the capacitors are fully charged.

Solution:

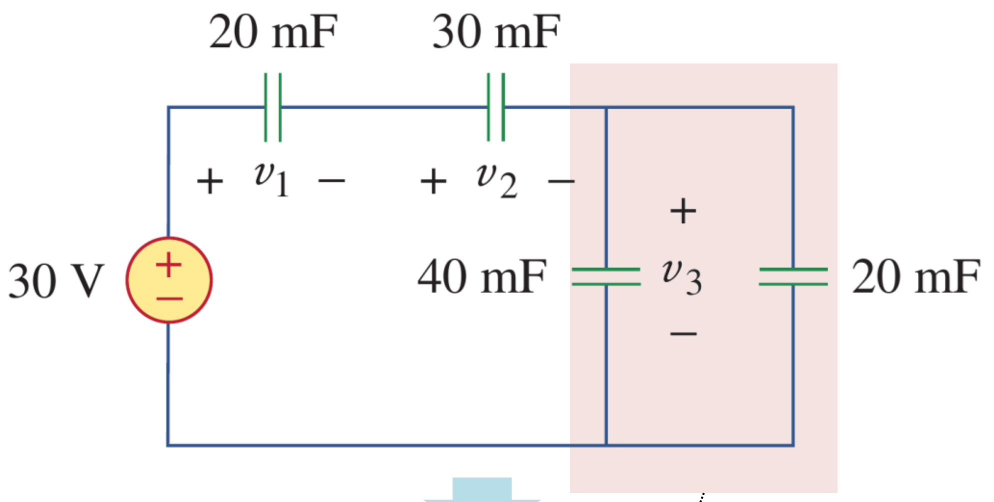
(1) Simplification of the circuit.

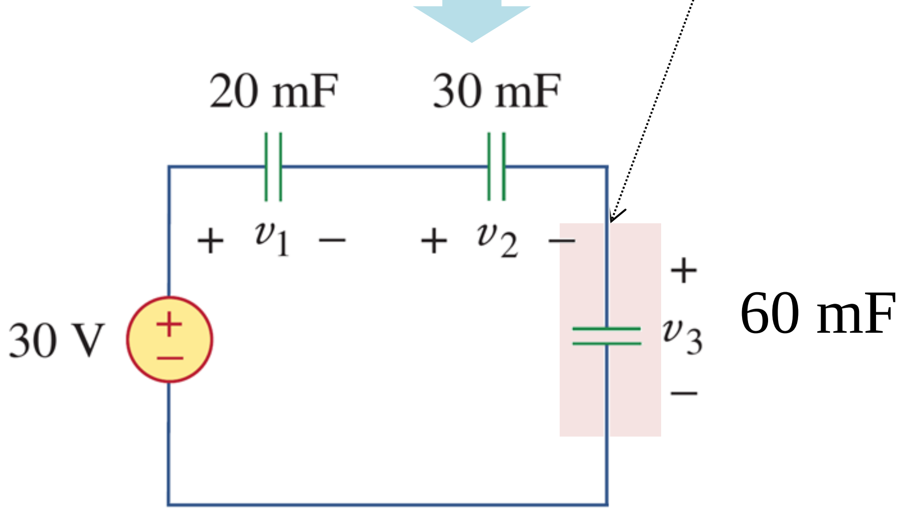
(2) Equivalent capacitance of all capacitors:

$$C_{eq} = \frac{1}{\frac{1}{60} + \frac{1}{30} + \frac{1}{20}} \text{ mF} = 10 \text{ mF}$$

(3) Total charge stored:

$$q = C_{eq}v = 10 \times 10^{-3} \times 30 = 0.3 \text{ C}$$

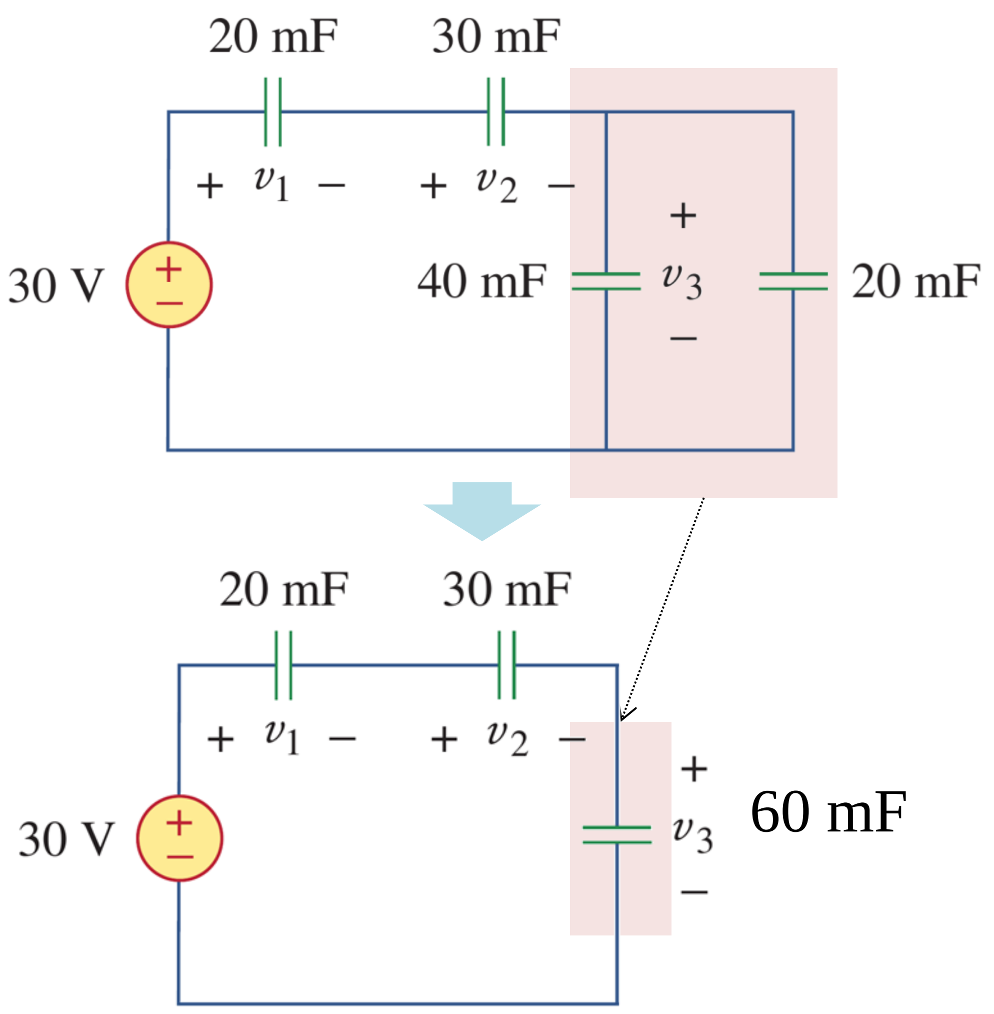




6.2 Capacitor networks (e)

- Example 6.05

Example 6.05: Find the voltage across each capacitor when the capacitors are fully charged.



Solution:

(4) Since 20 mF and 30 mF capacitors are in series:

$$v_1 = \frac{q}{C_1} = \frac{0.3}{20 \times 10^{-3}} = 15 \text{ V}$$

$$v_2 = \frac{q}{C_2} = \frac{0.3}{30 \times 10^{-3}} = 10 \text{ V}$$

$$v_3 = \frac{q}{60 \text{ mF}} = \frac{0.3}{60 \times 10^{-3}} = 5 \text{ V}$$

Lesson 6 Capacitors and inductors

6.3 Inductor basics (a)

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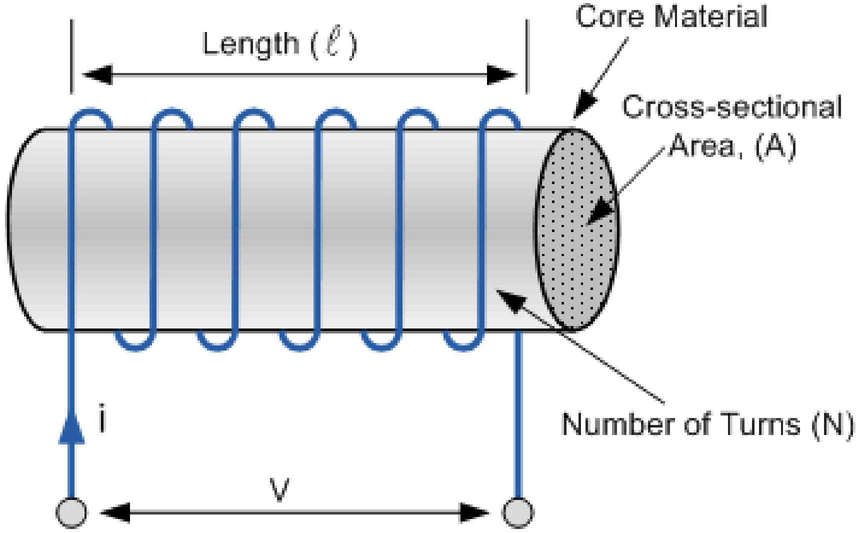
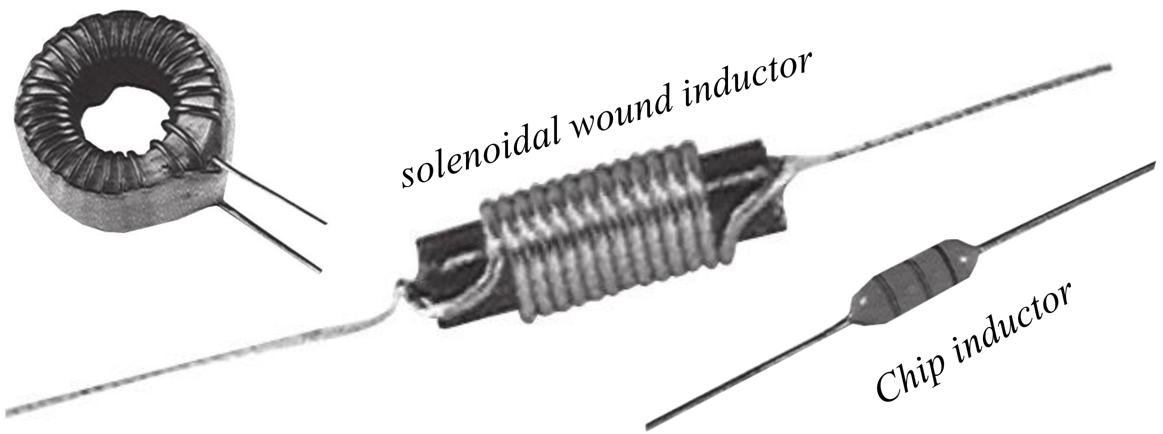
- Inductors:

a) An inductor (电感) is a passive element that stores energy in its magnetic field. It consists of conducting wires in forms of cylindrical coil and sometimes around a magnetic core (for magnetic enhancement).

b) Inductance is the property that an inductor exhibits opposition to the change of current flowing through it, measured in Henrys (H).





6.3 Inductor basics (b)

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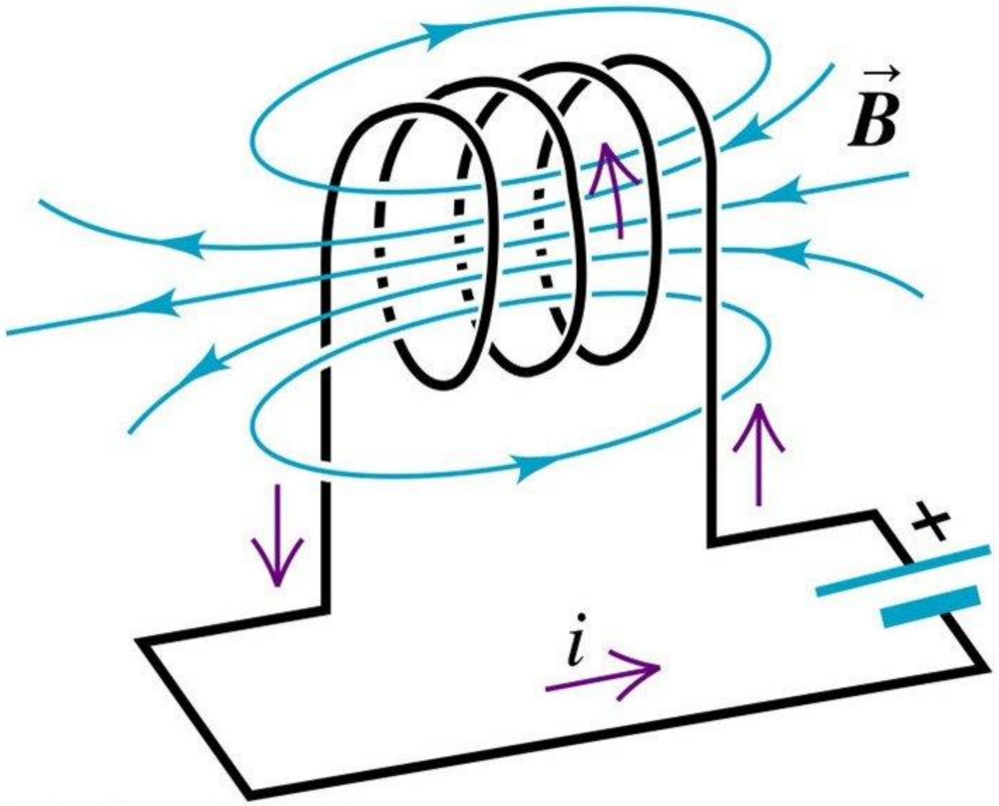
- Inductors:



An inductor resists the change in current.

Right hand grip rule

Faraday law of electromagnetic induction

Lenz's Law

Check more if you are interested

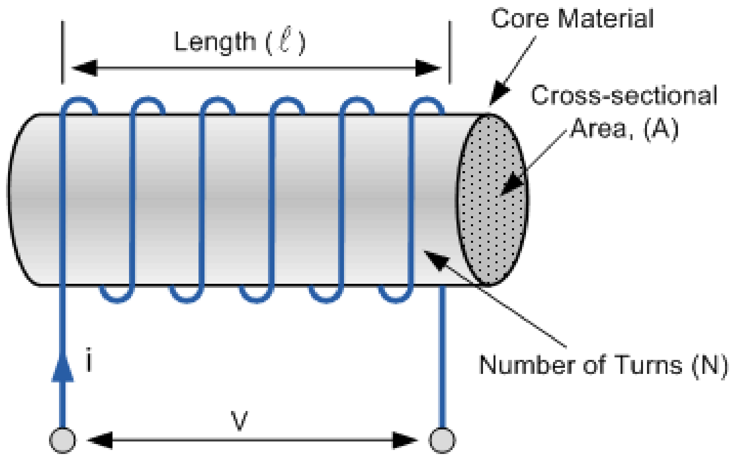
6.3 Inductor basics (c)

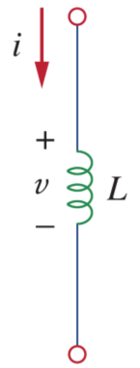
- Inductance:

The inductance of an inductor depends on its physical dimension and construction. One formula for calculating the inductance of inductors is shown below:

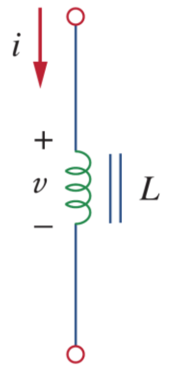
$$\boxed{L = \frac{N^2 \mu A}{l}}$$

where N is the number of turns, l is the length, A is the cross-sectional area, and μ is the permeability (磁导率) of the core.

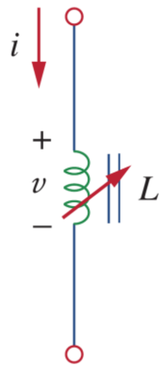




Air-core inductors



Iron-core inductors



Variable Iron-core inductors

6.3 Inductor basics (d)

- *v-i* characteristics of inductors:

a) Inductors are linear elements.

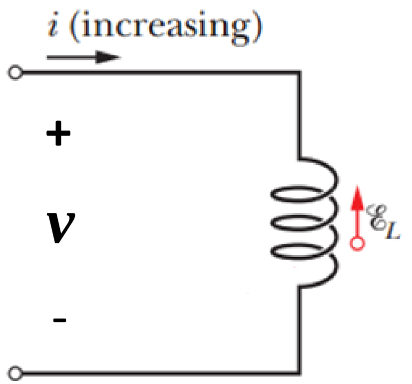
b) The *v-i* characteristics:

$$\boxed{v = L\frac{di}{dt}}$$

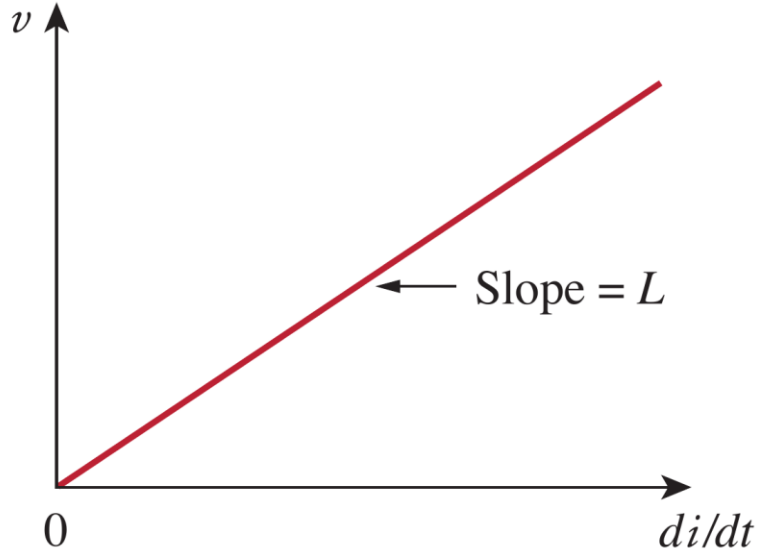
c) And:

$$\boxed{i(t) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)d\tau = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)}$$

which shows that Current (magnetic field) through an inductor reflects the history of the voltage applied onto it.



Passive sign convention



6.3 Inductor basics (e)

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- Power and energy:

a) Instant power (瞬态功率): $\boxed{p = vi = \left(L\frac{di}{dt}\right)i = iL\frac{di}{dt}}$

b) The energy stored in the inductor:

$$w = \int_{-\infty}^{t} p(\tau)d\tau = \int_{-\infty}^{t} Li'\frac{di'}{d\tau}d\tau = \int_{-\infty}^{i} Li'di' = \boxed{\frac{1}{2}Li^2}$$

in which $i(-\infty) = 0$, because the capacitor was uncharged at t = -∞.

This equation represents the energy stored in the magnetic field generated by the inductor. This energy can be retrieved later, since an ideal inductor cannot dissipate energy.

6.3 Inductor basics (g)

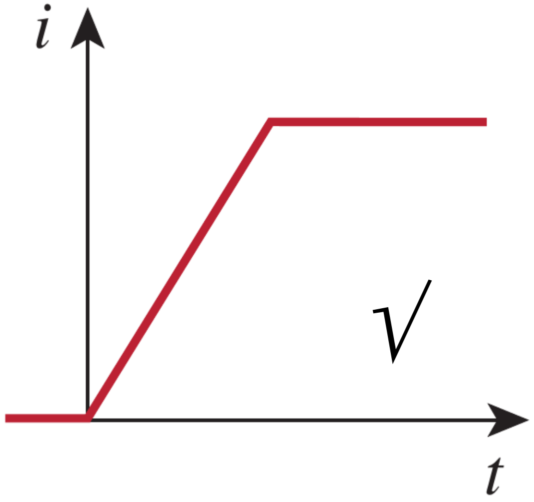
- Other important properties of inductors:

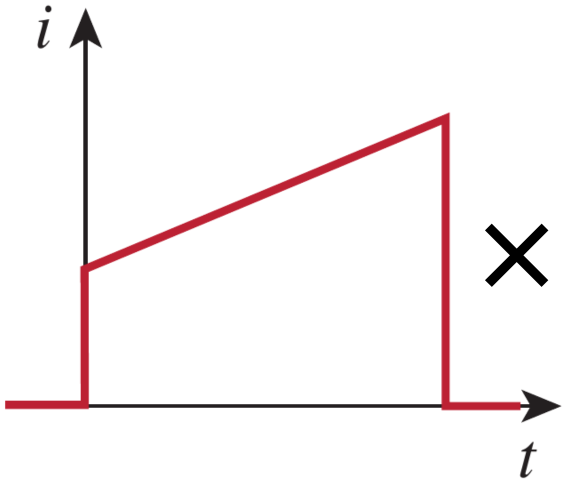
a) An inductor is an short circuit to DC (v = 0):

$$v = L\frac{di}{dt}, and\ \frac{di}{dt} = 0 \text{ for DC}$$

b) The current through a inductor cannot change abruptly:

A discontinuous change in the current through an inductor requires an infinite voltage, which is not physically possible. Thus, an inductor opposes an abrupt change in the current through it.





6.3 Inductor basics (h)

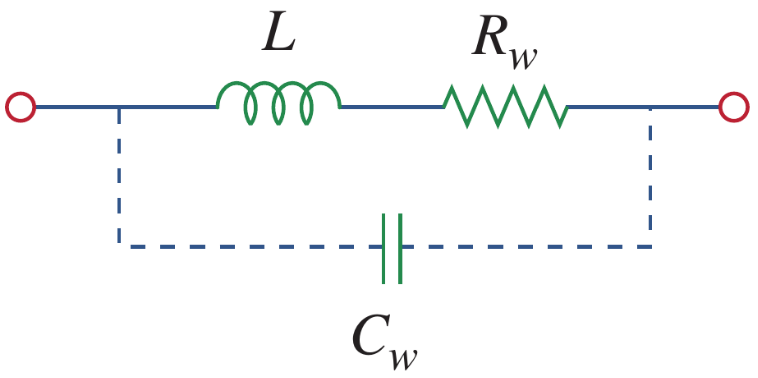
- Ideal and nonideal inductors

a) The ideal inductor does not dissipate energy. It takes power from the circuit when storing energy in its field and returns previously stored energy when delivering power to the circuit.

b) The equivalent circuit of a real, nonideal inductor has a winding resistance in series with an ideal inductor R_w, in addition to a parallel capacitor C_w. R_w results in power dissipation, but it is small and can be ignored in many cases. C_w is due to the capacitive coupling between the conducting coils, and is very small and can be ignored in most cases, except at high frequencies.

The equivalent circuit of an nonideal inductor

Winding resistance R_w
绕组电阻



Winding resistance C_w
绕组电容

- Example 6.06

Example 6.06: (a) If the current through a 0.1 H inductor is i(t) = 10te^{-5t} A, find the voltage across the inductor and the energy stored in it. (b) Find the current if the voltage across it is v(t) = 30t^2, assuming that the initial current is zero.

Solution: (a) $v = 0.1\frac{d}{dt}(10te^{-5t}) = e^{-5t} + t(-5)e^{-5t} = e^{-5t}(1 - 5t)$ V

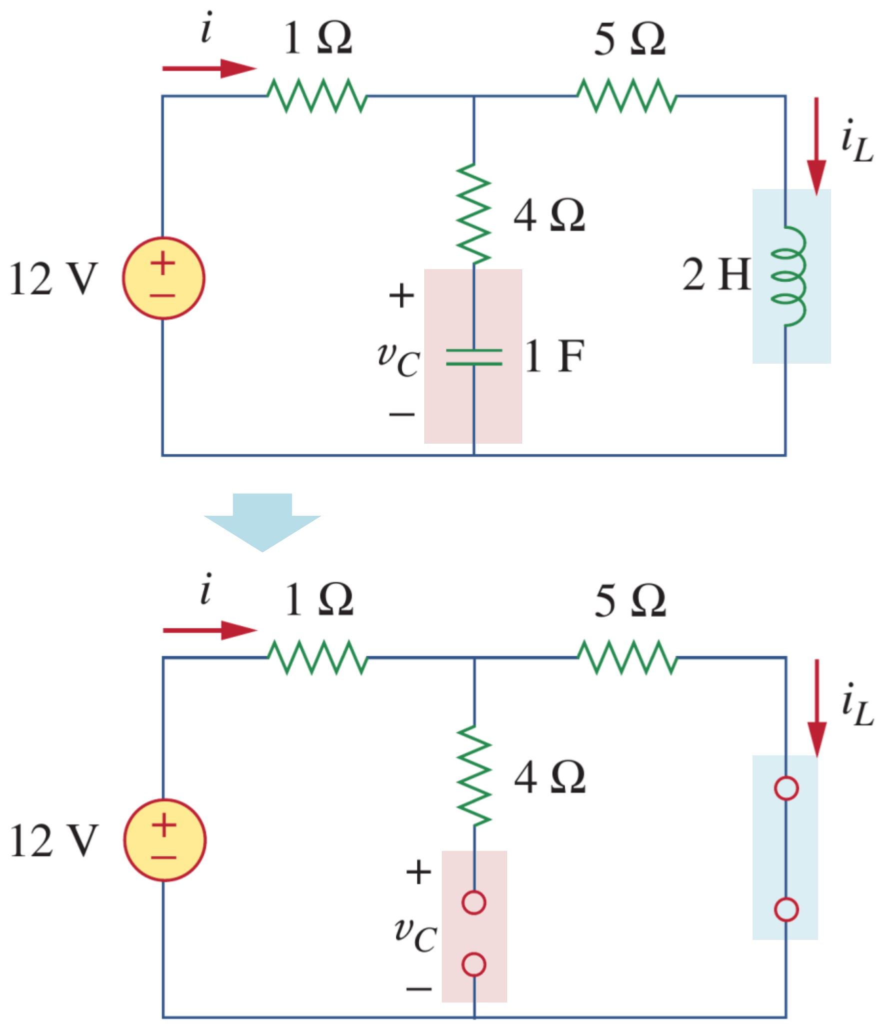
$$w = \frac{1}{2}Li^2 = \frac{1}{2}(0.1)100t^2e^{-10t} = 5t^2e^{-10t} \text{ J}$$

(b) $i = \frac{1}{L}\int_{t_0}^{t} v(t)dt + i(t0)\ and\ L = 0.1$ H

$$i = \frac{1}{0.1}\int_{0}^{t} 30t^2dt + 0 = 300 \times \frac{t^3}{3} = 100t^3 \text{ A}$$

- Example 6.07

Example 6.07: Under DC conditions, find: (a) i, v_C, and i_L, (b) the energy stored in the capacitor and inductor.



Solution:

$$i = i_L = \frac{12}{1+5} = 2\text{ A}$$

$$w_L = \frac{1}{2}Li_L^2 = \frac{1}{2}(2)(2^2) = 4\text{ J}$$

$$v_C = 5i = 10\text{ V}$$

$$w_C = \frac{1}{2}Cv_C^2 = \frac{1}{2}(1)(10^2) = 50\text{ J}$$

6.4 Inductor networks (a)

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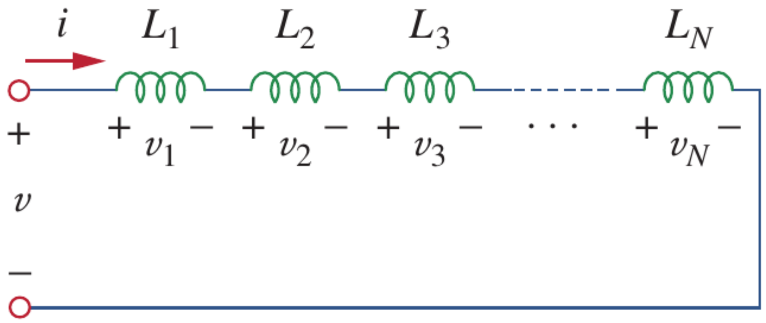
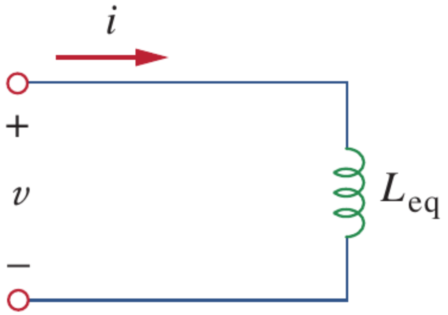
- Inductors in series (串联)





a) Applying KVL for the loop:

$$v = v_1 + v_2 + v_3 + \cdots + v_N$$

b) Since $v = L di/dt$, as:

$$v = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + L_3 \frac{di}{dt} + \cdots + L_N \frac{di}{dt} = \left(\sum_{k=1}^{N} L_k \right) \frac{di}{dt} = L_{eq} \frac{di}{dt}$$

c) For series inductors, the equivalent inductance is:

$$\boxed{L = L_1 + L_2 + L_3 + \cdots + L_N}$$

The same formula as for the resistance R.

6.4 Inductor networks (b)

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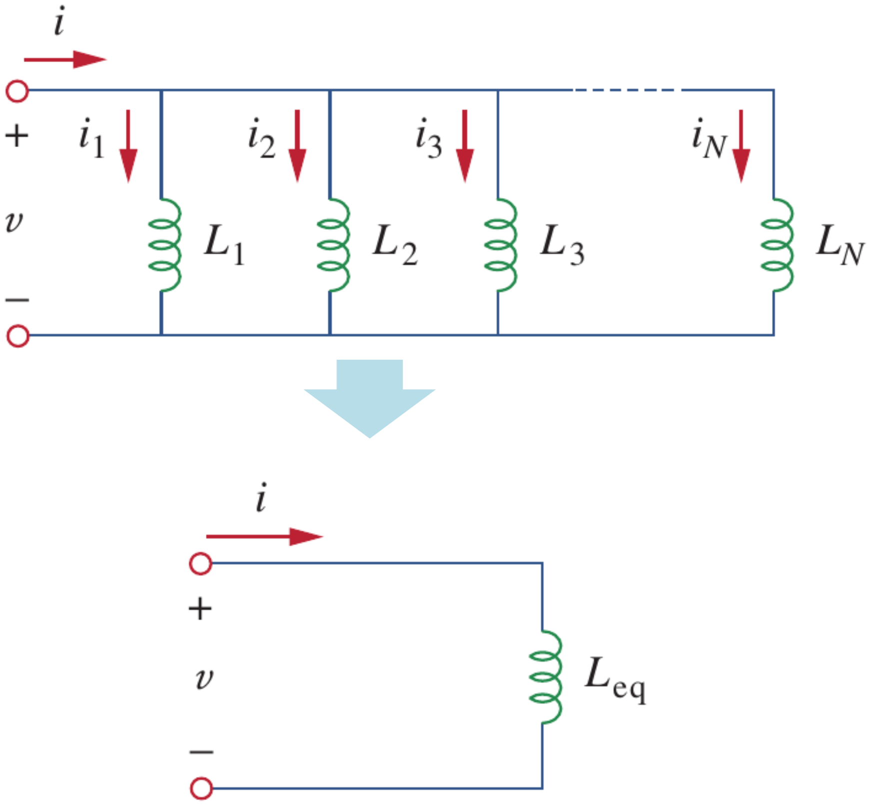
- Inductors in parallel（并联）



a) Applying KCL at the top node: $i = i_1 + i_2 + i_3 + \cdots + i_N$

b) Since $i(t) = \dfrac{1}{L}\displaystyle\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$

$$i = \left(\sum_{k=1}^{N}\frac{1}{L_k}\right)\int_{t_0}^{t} vdt + \sum_{k=1}^{N} i_k(t_0) = \frac{1}{L_{eq}}\int_{t_0}^{t} vdt + i(t_0)$$

c) So: $\boxed{\dfrac{1}{L_{eq}} = \dfrac{1}{L_1} + \dfrac{1}{L_2} + \dfrac{1}{L_3} + \cdots + \dfrac{1}{L_N}}$

6.4 Inductor networks (c)

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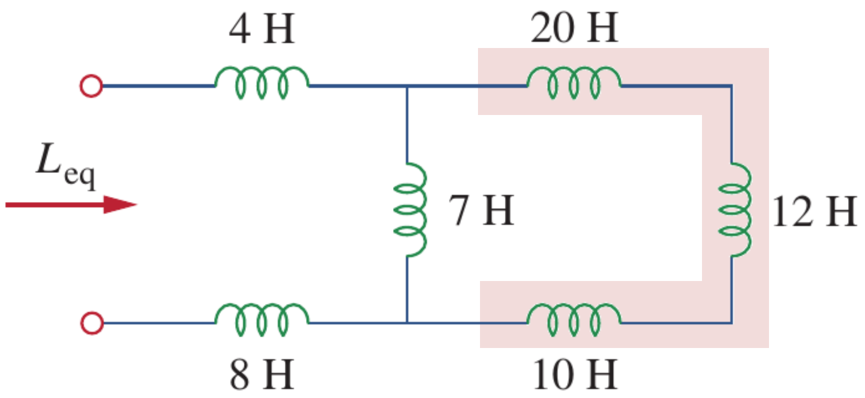
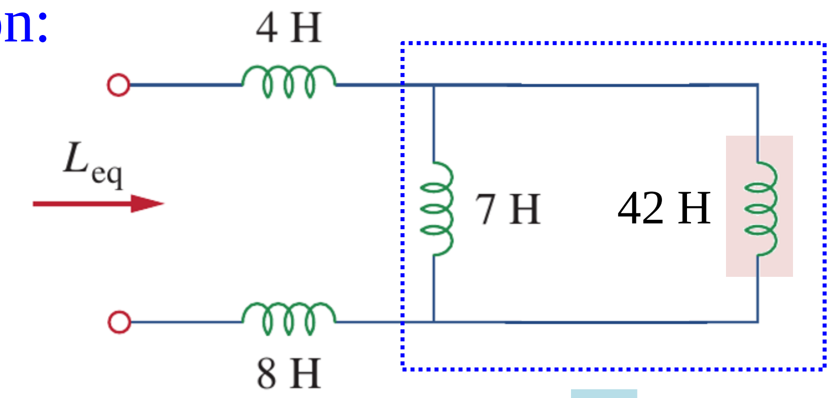
- Example 6.08

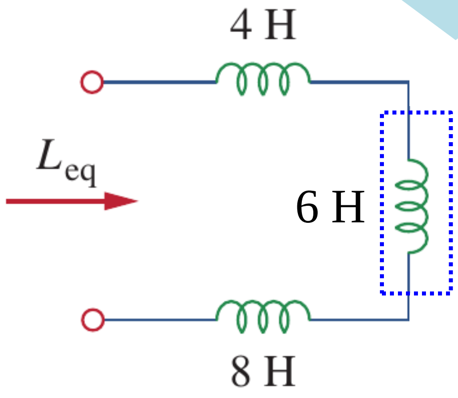
Example 6.08: Find the equivalent inductance of the following circuit



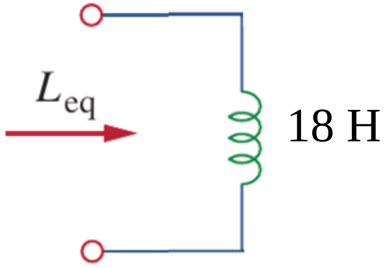
Solution:



$$L'_{eq} = 20 + 12 + 10$$
$$= 42 \text{ H}$$



$$1/L''_{eq} = 1/7 + 1/42$$
$$L''_{eq} = \frac{7 \times 42}{7 + 42} = 6 \text{ H}$$



$$L_{eq} = 4 + 6 + 8$$
$$= 18\ H$$

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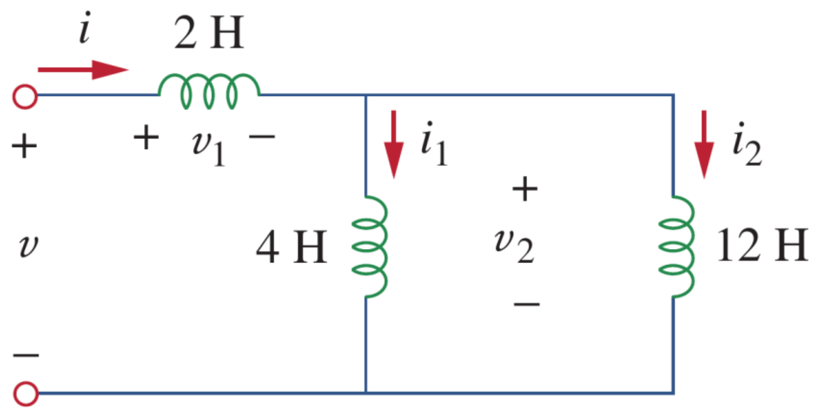
- Example 6.09

Example 6.09: For the following circuit, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find:
(a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$;
(c) $i_1(t)$ and $i_2(t)$.



Solution: (a) to find $i_1(0)$

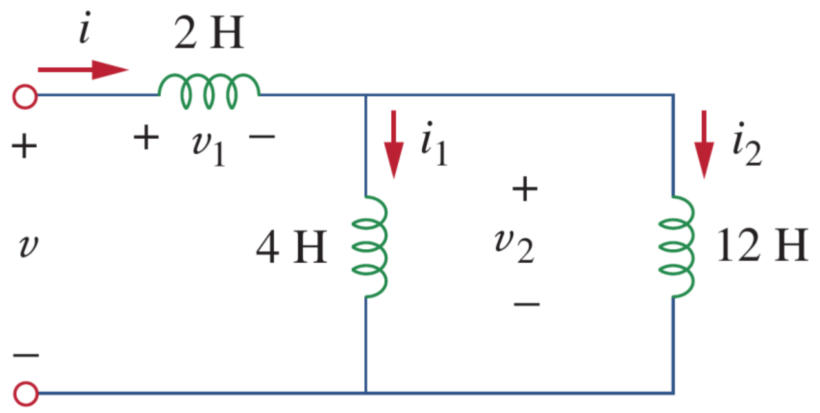
(a) From $i(t) = 4(2 - e^{-10t})$ mA, $i(0) = 4(2 - 1) = 4$ mA. Since $i = i_1 + i_2$,

$$i_1(0) = i(0) - i_2(0) = 4 - (-1) = 5 \text{ mA}$$

6.4 Inductor networks (e)

- Example 6.09

Example 6.09: For the following circuit, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find:
(a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$;
(c) $i_1(t)$ and $i_2(t)$.



Solution: (b) to find $v(t)$, $v_1(t)$ and $v_2(t)$

The equivalent inductan $L_{eq} = 2 + 4 \| 12 = 2 + 3 = 5$ H

$$v(t) = L_{eq}\frac{di}{dt} = 5(4)(-1)(-10)e^{-10t} \text{ mV} = 200e^{-10t} \text{ mV}$$

$$v_1(t) = 2\frac{di}{dt} = 2(-4)(-10)e^{-10t} \text{ mV} = 80e^{-10t} \text{ mV}$$

Since $v = v_1 + v_2$, $\quad v_2(t) = v(t) - v_1(t) = 120e^{-10t}$ mV

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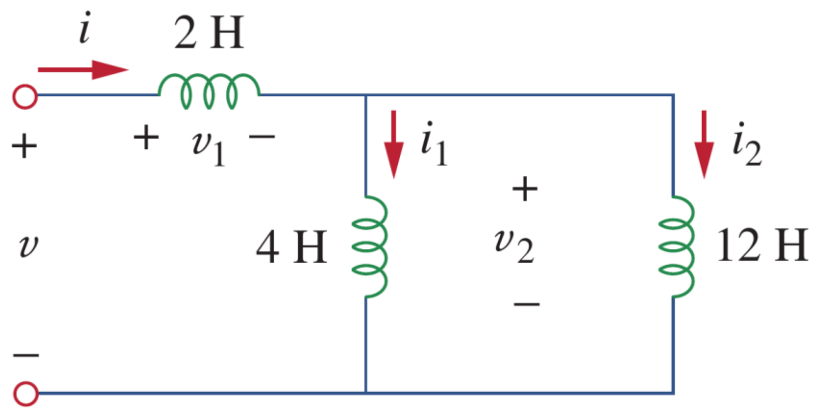
- Example 6.09

Example 6.09: For the following circuit, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find:
(a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$;
(c) $i_1(t)$ and $i_2(t)$.



Solution: (c) to find $i_1(t)$ and $i_2(t)$

$$i_1(t) = \frac{1}{4}\int_0^t v_2\,dt + i_1(0) = \frac{120}{4}\int_0^t e^{-10t}\,dt + 5\text{ mA}$$

$$= -3e^{-10t}\Big|_0^t + 5\text{ mA} = -3e^{-10t} + 3 + 5 = 8 - 3e^{-10t}\text{ mA}$$

$$i_2(t) = \frac{1}{12}\int_0^t v_2\,dt + i_2(0) = \frac{120}{12}\int_0^t e^{-10t}\,dt - 1\text{ mA}$$

$$= -e^{-10t}\Big|_0^t - 1\text{ mA} = -e^{-10t} + 1 - 1 = -e^{-10t}\text{ mA}$$

Lesson 6 Capacitors and inductors

6.5 Resistors, capacitors and inductors (a)

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- Comparison

a) Resistors, capacitors and inductors are all linear passive elements.

b) Passive sign convention is assumed.

c) Ideal resistors consume energy, while ideal capacitors and inductors store energy without energy dissipation.

d) The Δ - Y transformation can also be applied to inductors and capacitors in a similar manner, as long as all elements are the same type.

TABLE 6.1

Important characteristics of the basic elements.†

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v-i:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$	$v = L\frac{di}{dt}$
i-v:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$
p or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

6.5 Resistors, capacitors and inductors (b)

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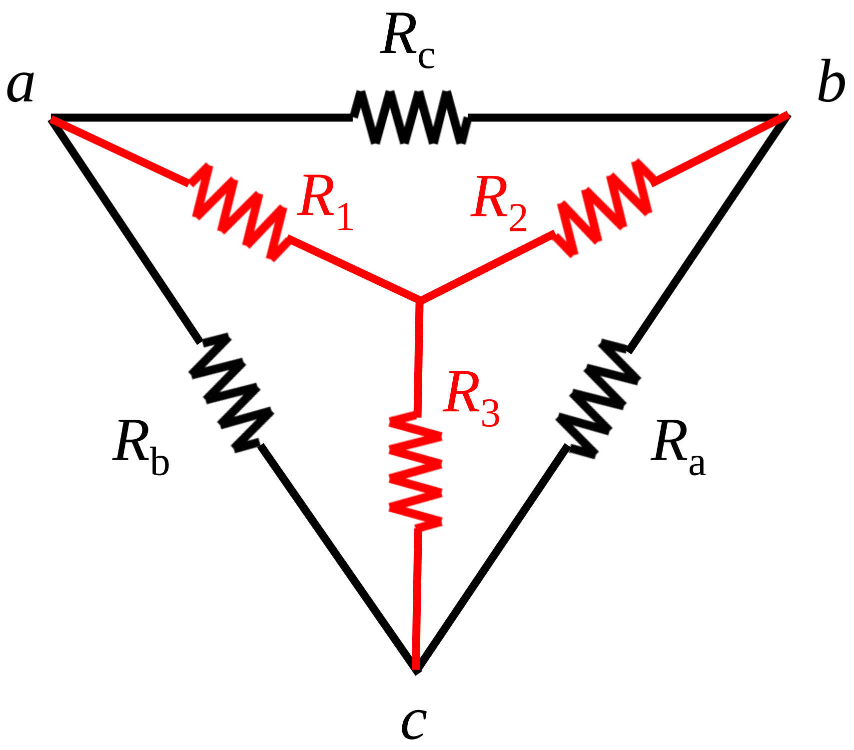
- Δ – Y transformation for **Resistors**

Δ to Y

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_b R_a}{R_a + R_b + R_c}$$



Y to Δ

$$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$$

6.5 Resistors, capacitors and inductors (c)

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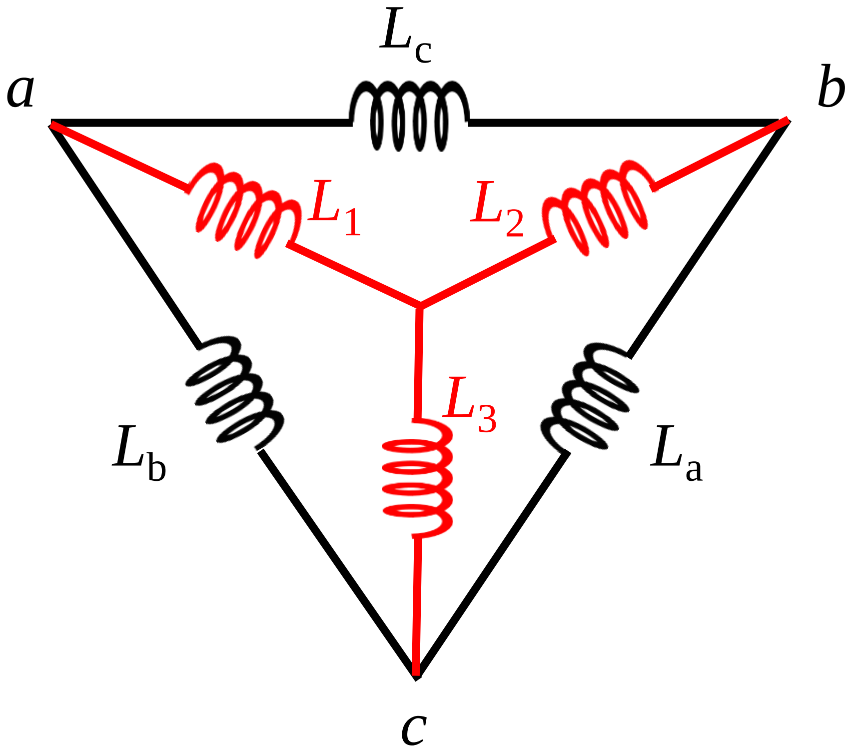
- Δ – Y transformation for **Inductors**

Δ to Y

$$L_1 = \frac{L_b L_c}{L_a + L_b + L_c}$$

$$L_2 = \frac{L_a L_c}{L_a + L_b + L_c}$$

$$L_3 = \frac{L_b L_a}{L_a + L_b + L_c}$$



Y to Δ

$$L_a = \frac{L_1 L_2 + L_1 L_3 + L_2 L_3}{R_1}$$

$$L_b = \frac{L_1 L_2 + L_1 L_3 + L_2 L_3}{R_2}$$

$$L_c = \frac{L_1 L_2 + L_1 L_3 + L_2 L_3}{R_3}$$

6.5 Resistors, capacitors and inductors (d)

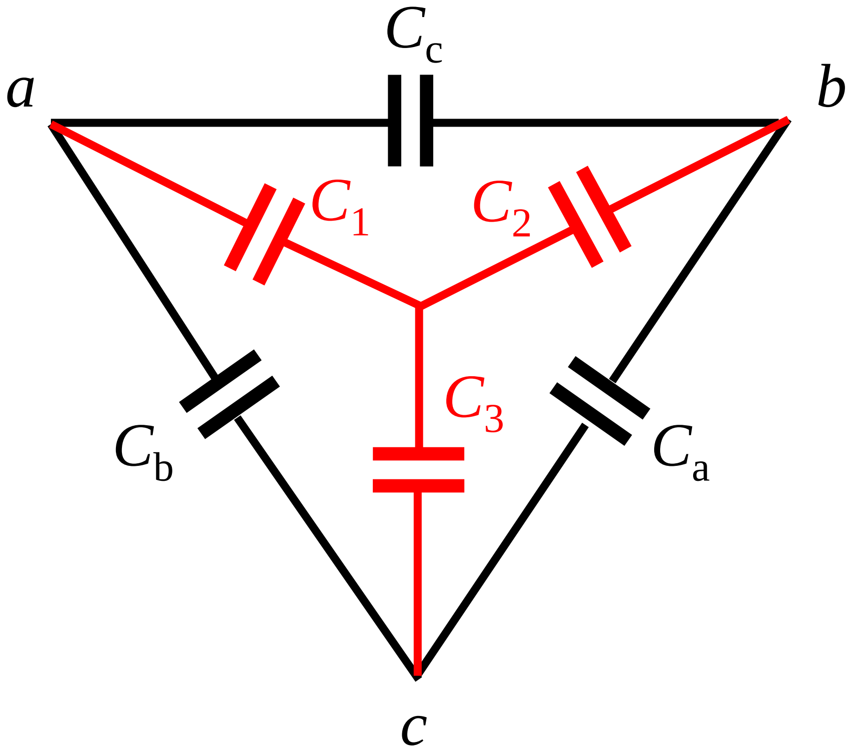
- Δ – Y transformation for **Capacitors**

Δ to Y

$$C_1 = \frac{C_aC_b + C_bC_c + C_aC_c}{C_a}$$

$$C_2 = \frac{C_aC_b + C_bC_c + C_aC_c}{C_b}$$

$$C_3 = \frac{C_aC_b + C_bC_c + C_aC_c}{C_c}$$



Y to Δ

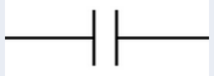
$$C_a = \frac{C_2C_3}{C_1 + C_2 + C_3}$$

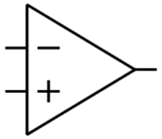
$$C_b = \frac{C_1C_3}{C_1 + C_2 + C_3}$$

$$C_c = \frac{C_1C_2}{C_1 + C_2 + C_3}$$

6.5 Resistors, capacitors and inductors (e)

- Important elements learned so far

Type	symbol	Calculation in Series	Calculation in Parallel	Continuity
Resistor 电阻		Direct sum	Reciprocal sum	
Inductor 电感		Direct Sum	Reciprocal Sum	Current (magnetic flux)
Conductor 电导		Reciprocal Sum	Direct sum	
Capacitor 电容		Reciprocal Sum	Direct sum	Voltage (charge)

Op amp 运放 

Virtual shorting (虚短), virtual opening (虚断)

Lesson 6 Capacitors and inductors

6.6 Integrators and differentiators (a)

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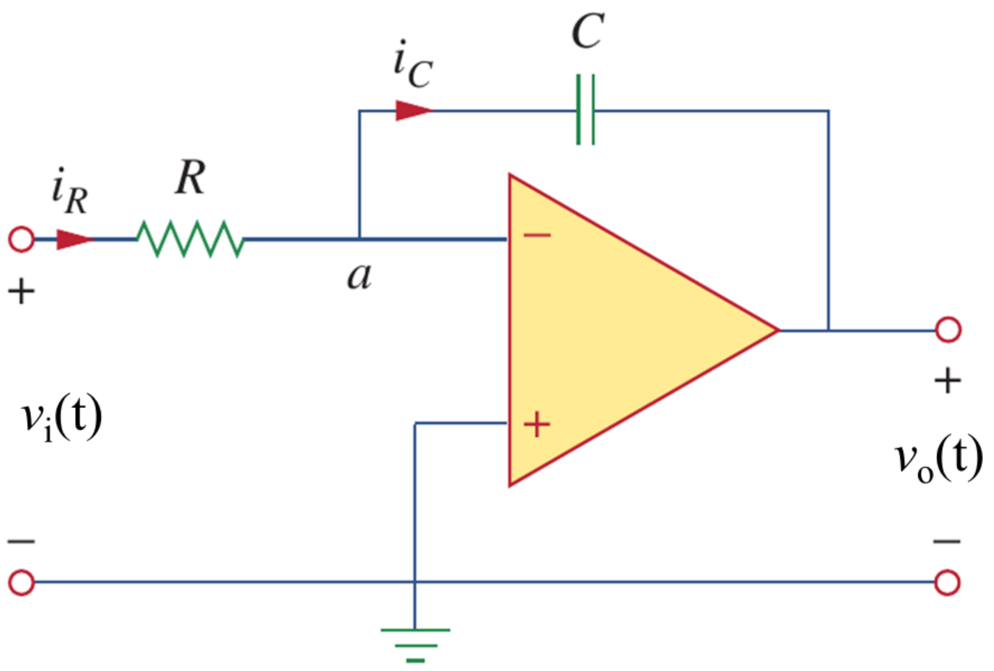
- Op-amps with capacitors: integrators



a) An integrator is an op amp circuit whose output is proportional to the integral of the input signal.

b) Since $i_+ = i_- = 0$ (虚断): $\dfrac{v_a - v_i}{R} = C\dfrac{d(v_o - v_a)}{dt}$

c) Since $v_+ = v_-$ and $v_+ = 0$ (虚短), $v_a = 0$:

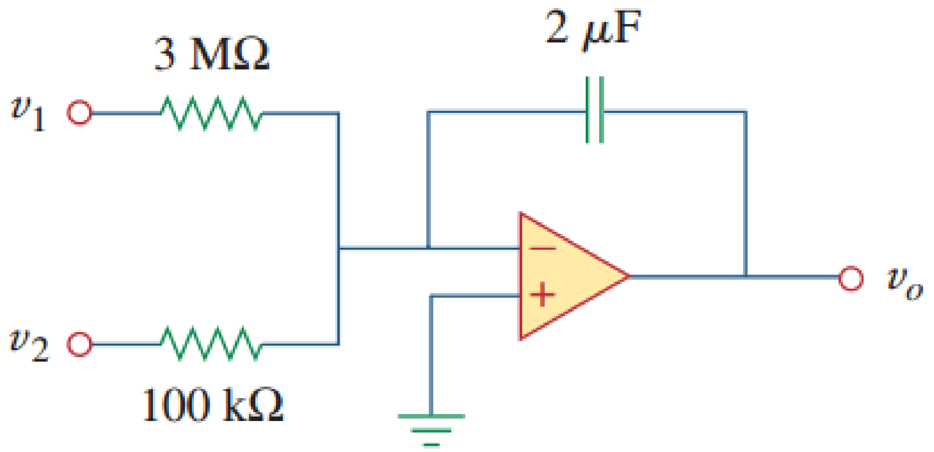
$$\frac{-v_i}{R} = C\frac{dv_o}{dt} \implies dv_o = -\frac{1}{RC}v_i dt$$

d) Therefore: $v_o(t) = -\dfrac{1}{RC}\displaystyle\int_0^t v_i dt + v_o(0)$

If $v_o(0) = 0$ V, $v_o(t) = -\dfrac{1}{RC}\displaystyle\int_0^t v_i dt$

- Example 6.10

Example 6.10: If v_1 = 10cos2t mV and v_2 = 0.5t mV, find v_o in the op amp circuit. Assume that the voltage across the capacitor is initially zero.



Solution:

$$v_o = -\frac{1}{R_1C}\int v_1\,dt - \frac{1}{R_2C}\int v_2\,dt$$

$$= -\frac{1}{3\times10^6\times2\times10^{-6}}\int_0^t 10\cos(2\tau)d\tau$$

$$-\frac{1}{100\times10^3\times2\times10^{-6}}\int_0^t 0.5\tau\,d\tau$$

$$= -\frac{1}{6}\frac{10}{2}\sin 2t - \frac{1}{0.2}\frac{0.5t^2}{2} = -0.833\sin 2t - 1.25t^2\ \text{mV}$$

6.6 Integrators and differentiators (c)

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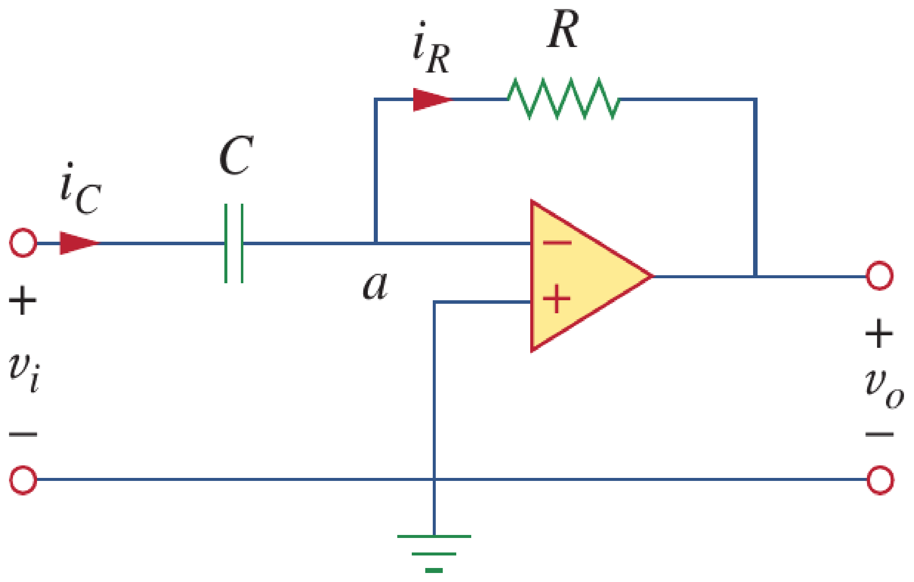
- Op-amps with capacitors: differentiators



a) A differentiator is an op amp circuit whose output is proportional to the rate of change of the input signal.

b) Since $i_+ = i_- = 0$ (虚断): $C\dfrac{d(v_a - v_i)}{dt} = \dfrac{v_o - v_a}{R}$

c) Since $v_+ = v_-$ and $v_+ = 0$ (虚短), $v_a = 0$:

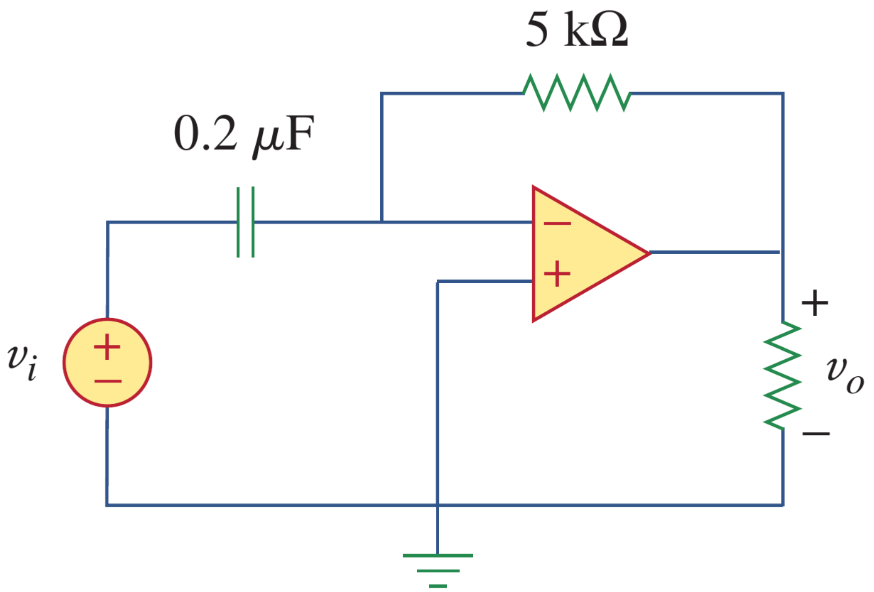
$$-C\frac{d(v_i)}{dt} = \frac{v_o}{R}$$

d) Therefore:

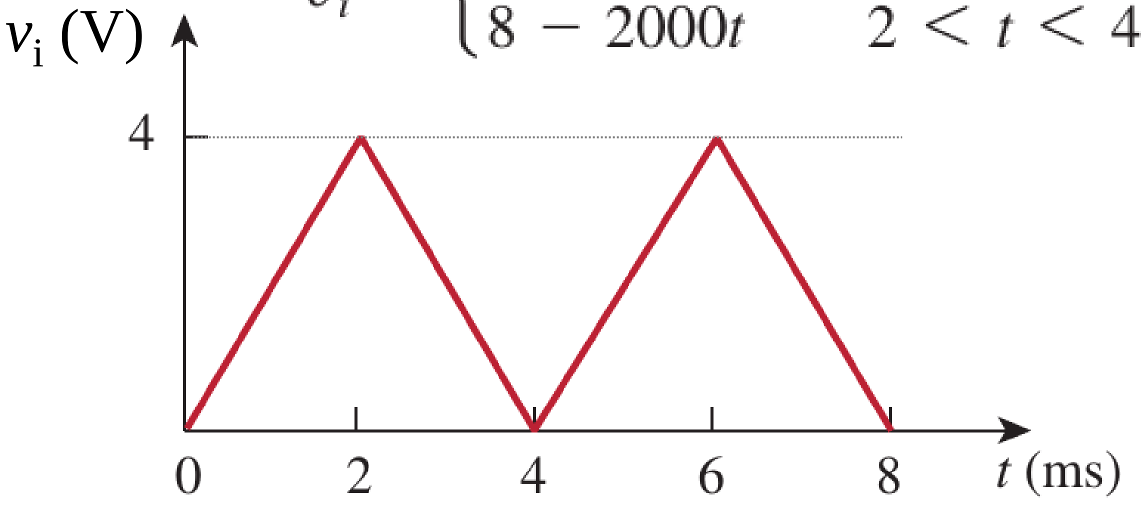
$$v_o = -RC\frac{dv_i}{dt}$$

- Example 6.11

Example 6.11: Sketch the output voltage for the circuit, given the input voltage in the figure. Take $v_o = 0$ at t = 0.



$$v_i = \begin{cases} 2000t & 0 < t < 2 \text{ ms} \\ 8 - 2000t & 2 < t < 4 \text{ ms} \end{cases}$$



6.6 Integrators and differentiators (e)

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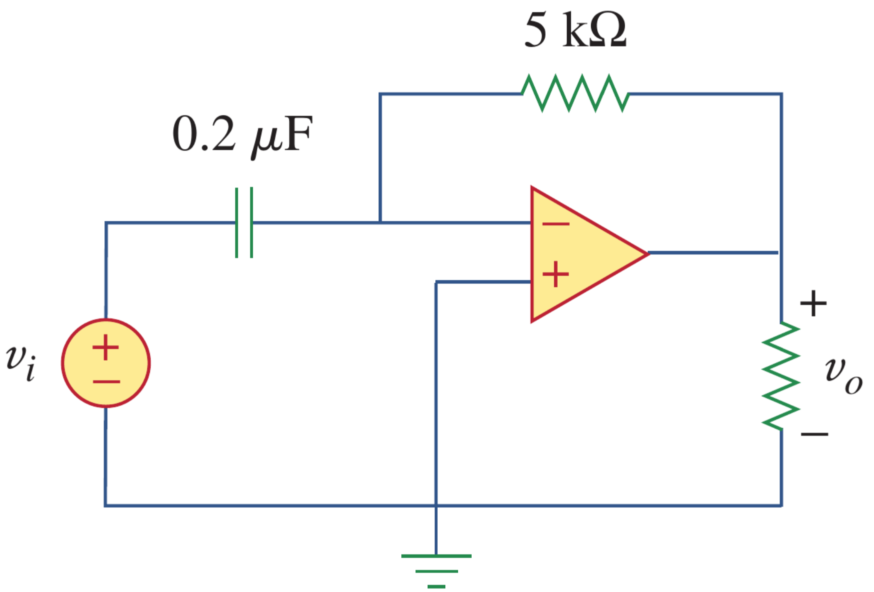
- Example 6.11

Example 6.11: Sketch the output voltage for the circuit, given the input voltage in the figure. Take $v_o = 0$ at t = 0.



Solution:

$$v_i = \begin{cases} 2000t & 0 < t < 2 \text{ ms} \\ 8 - 2000t & 2 < t < 4 \text{ ms} \end{cases}$$

$$v_o = -RC\frac{dv_i}{dt} = \begin{cases} -2 \text{ V} & 0 < t < 2 \text{ ms} \\ 2 \text{ V} & 2 < t < 4 \text{ ms} \end{cases}$$

$$RC = 5 \times 10^3 \times 0.2 \times 10^{-6} = 10^{-3} \text{ s}$$

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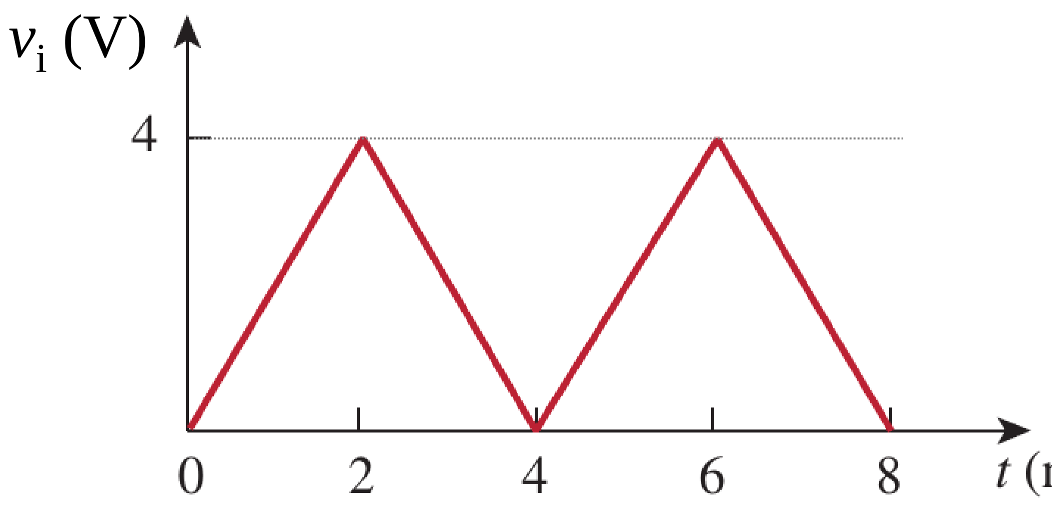
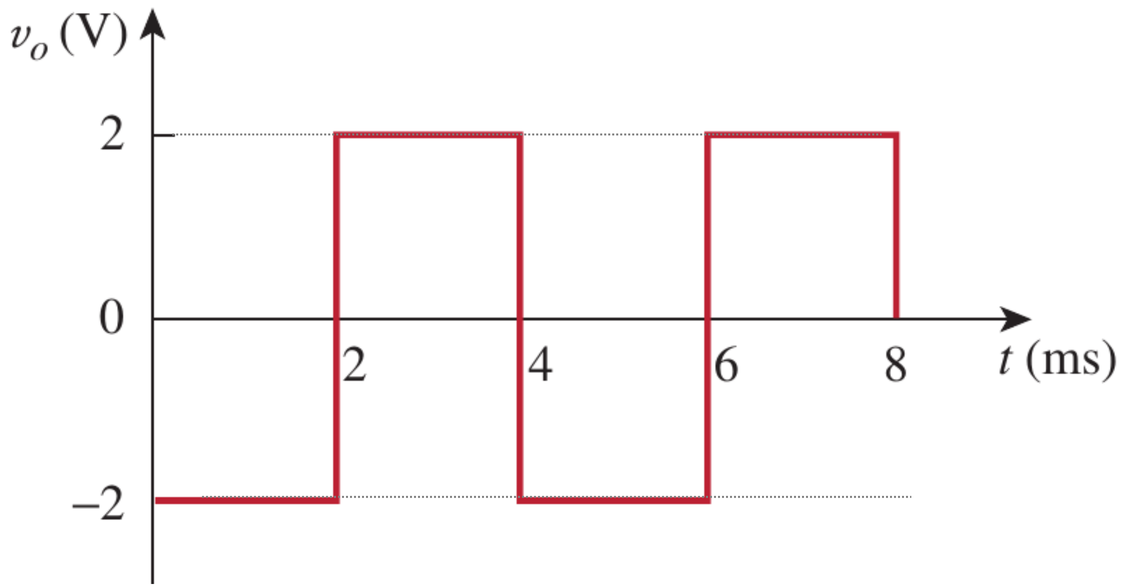
- Example 6.11

Example 6.11: Sketch the output voltage for the circuit, given the input voltage in the figure. Take $v_o = 0$ at t = 0.



Solution:



Lesson 6 Capacitors and inductors

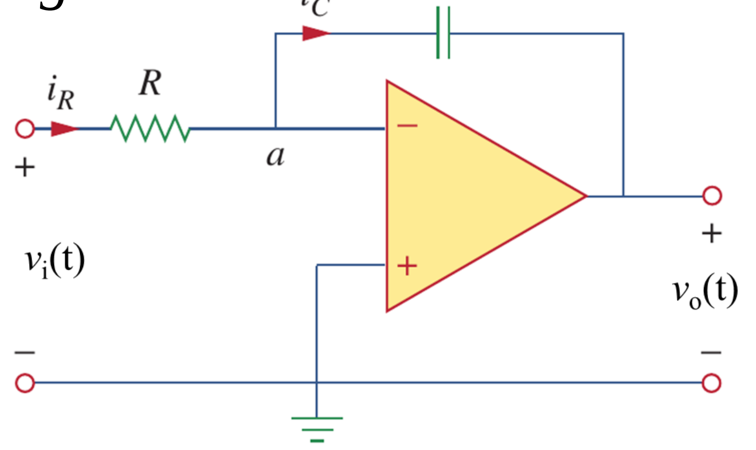
6.7 Summary and assignments

TABLE 6.1

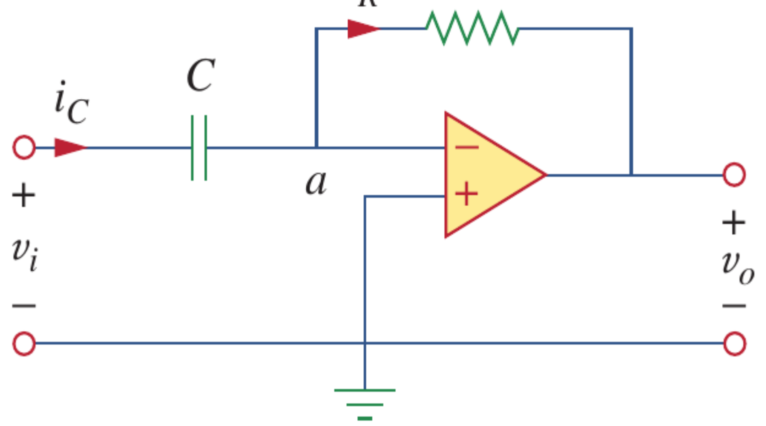
Important characteristics of the basic elements.†

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v-i:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$	$v = L\frac{di}{dt}$
i-v:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$
p or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

Integrator:



Differentiator:



6.7 Summary and assignments

Chapter 6:

Problems 6.13, 6.19, 6.22, 6.32, 6.55, 6.56,

(a) 6.59, 6.65, 6.72, 6.77, try these four if you get time.

(b) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.

(c) There are plenty of problems in the textbook, try to complete as many as you can.

Mammatus Clouds